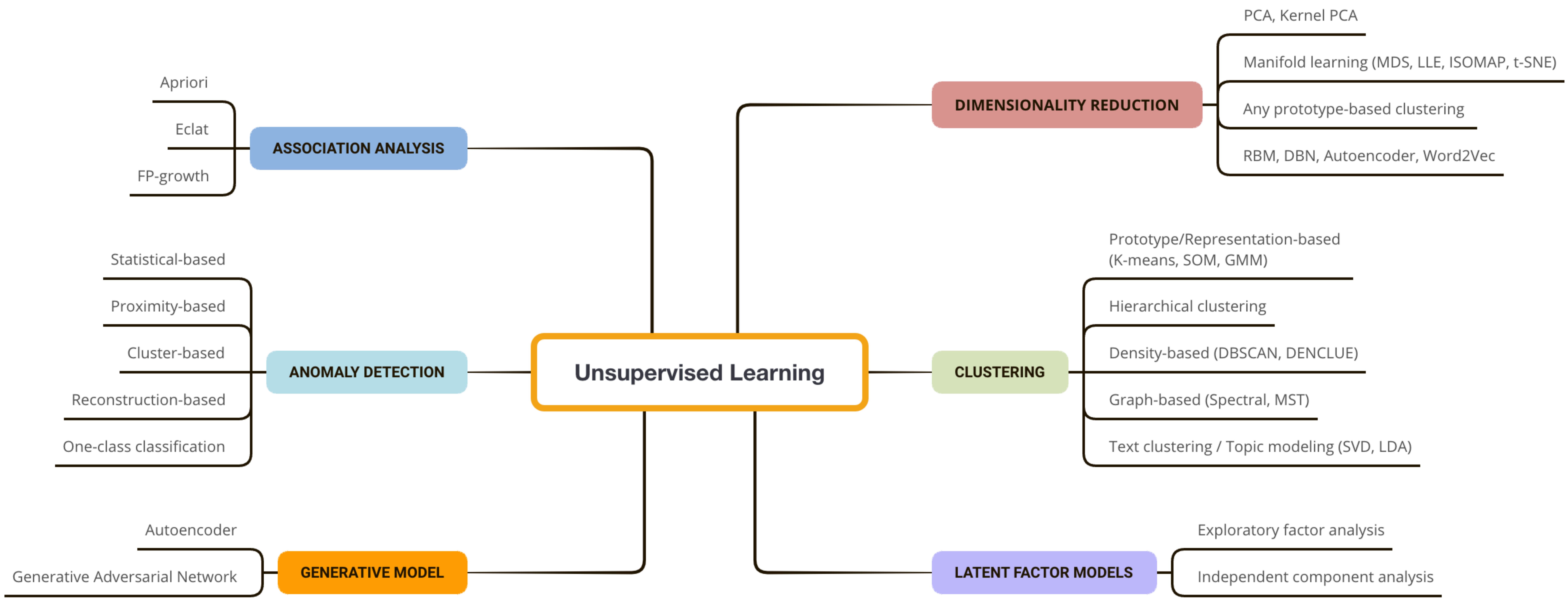
Unsupervised Learning


Unsupervised Learning
ASSOCIATION ANALYSIS
Apriori
Eclat
FP-growth
ANOMALY DETECTION
Statistical-based
Proximity-based
Cluster-based
Reconstruction-based
One-class classification
GENERATIVE MODEL
Autoencoder
Generative Adversarial Network
DIMENSIONALITY REDUCTION
PCA, Kernel PCA
Manifold learning (MDS, LLE, ISOMAP, t-SNE)
Any prototype-based clustering
RBM, DBN, Autoencoder, Word2Vec
CLUSTERING
Prototype/Representation-based (K-means, SOM, GMM)
Hierarchical clustering
Density-based (DBSCAN, DENCLUE)
Graph-based (Spectral, MST)
Text clustering / Topic modeling (SVD, LDA)
LATENT FACTOR MODELS
Exploratory factor analysis
Independent component analysis

Dimensionality Reduction

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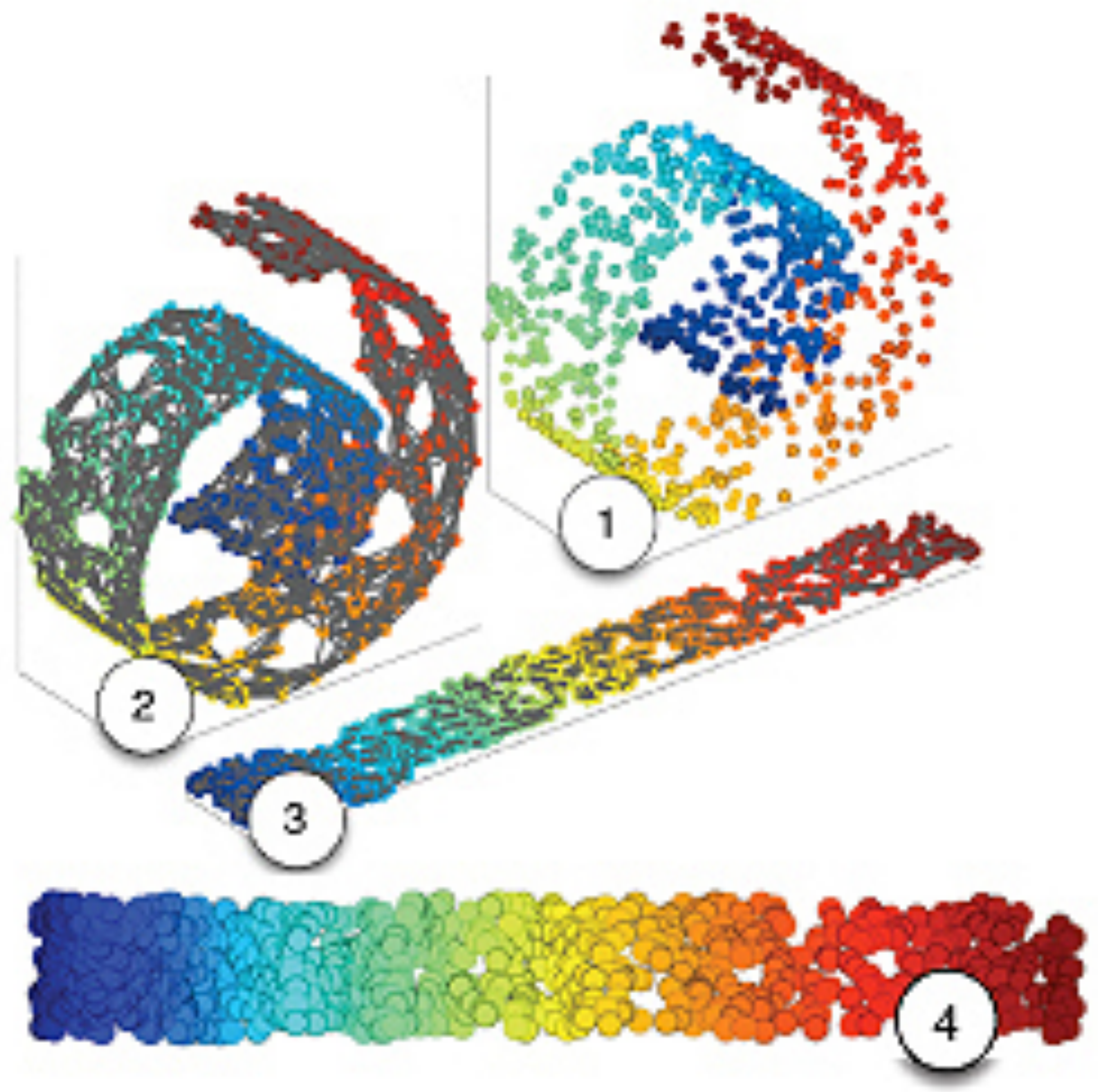
Machine Learning

Topics

- ☐ Principal component analysis (PCA)
- ☐ Manifold learning



Many Machine Learning problems involve thousands or even millions of features for each training instance. Not only does this make training extremely slow, it can also make it much harder to find a good solution, as we will see. This problem is often referred to as the ***curse of dimensionality***.

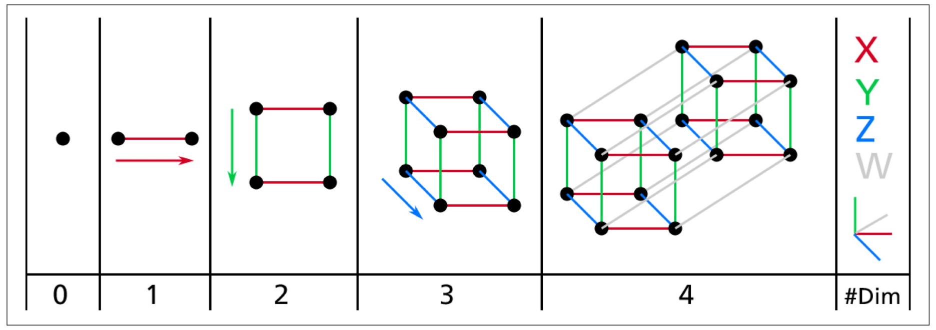


Figure 8-1. Point, segment, square, cube, and tesseract (0D to 4D hypercubes)[2]

Here is a more troublesome difference: if you pick two points randomly in a unit square, the distance between these two points will be, on average, roughly 0.52. If you pick two random points in a unit 3D cube, the average distance will be roughly 0.66. But what about two points picked randomly in a 1,000,000-dimensional hypercube? Well, the average distance, believe it or not, will be about 408.25 (roughly $\sqrt{1,000,000/6}$)! This is quite counterintuitive: how can two points be so far apart when they both lie within the same unit hypercube? This fact implies that high-dimensional datasets are at risk of being very sparse: most training instances are likely to be far away from each other. Of course, this also means that a new instance will likely be far away from any training instance, making predictions much less reliable than in lower dimensions, since they will be based on much larger extrapolations. In short, the more dimensions the training set has, the greater the risk of overfitting it.

Main Approaches for Dimensionality Reduction

1) Projection

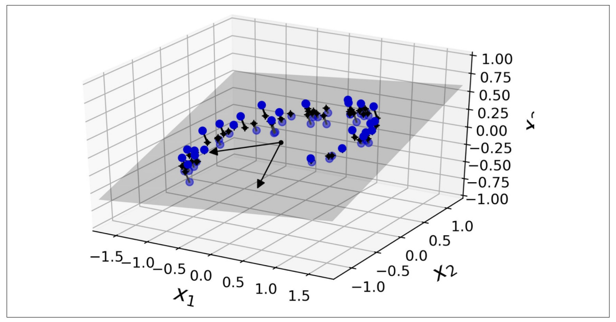


Figure 8-2. A 3D dataset lying close to a 2D subspace

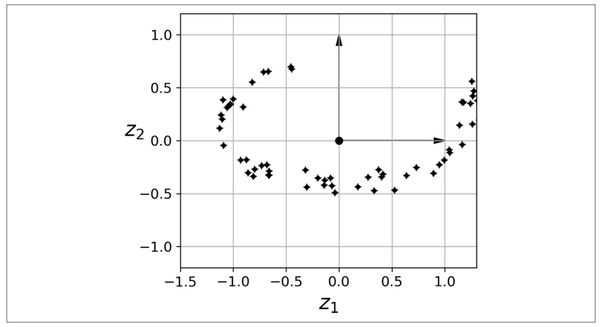
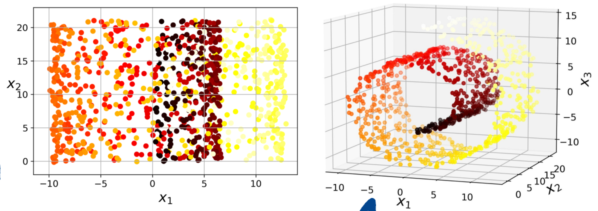


Figure 8-3. The new 2D dataset after projection

2) Manifold Learning



The Swiss roll is an example of a 2D *manifold*. Put simply, a 2D manifold is a 2D shape that can be bent and twisted in a higher-dimensional space. More generally, a d-dimensional manifold is a part of an n-dimensional space (where $d < n$) that locally resembles a d-dimensional hyperplane. In the case of the Swiss roll, $d = 2$ and $n = 3$: it locally resembles a 2D plane, but it is rolled in the third dimension.

Many dimensionality reduction algorithms work by modeling the *manifold* on which the training instances lie; this is called *Manifold Learning*. It relies on the *manifold assumption*, also called the *manifold hypothesis*, which holds that most real-world high-dimensional datasets lie close to a much lower-dimensional manifold. This assumption is very often empirically observed.

ตั้งสมมติฐานว่า high-dimensional dataset เป็น lower-dimensional manifold

เราจึงต้อง unrolled มันออกมา

✅ **Works Well**

When the lower-dimensional manifold truly simplifies the problem.

Example: On the Swiss roll, unrolling it makes two classes separable by a straight line.

❌ **Doesn't Always Help**

Sometimes, reducing dimensions can make patterns more complicated.

Example: If your decision boundary is simple in 3D (like a flat plane), it may look messy when flattened into 2D.

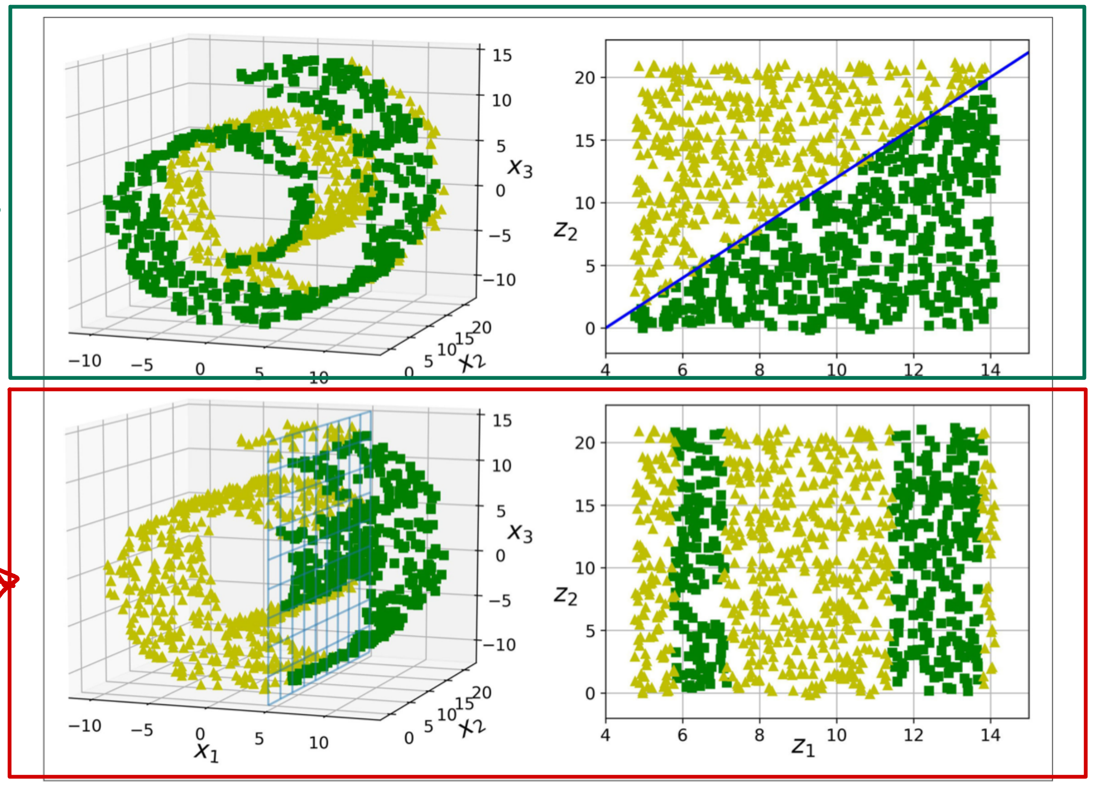


Figure 8-6. The decision boundary may not always be simpler with lower dimensions

Dimetional Reduction Algorithm

PCA

Principal Component Analysis (PCA) is by far the most popular dimensionality reduction algorithm. First it identifies the hyperplane that lies closest to the data, and then it projects the data onto it, just like in Figure 8-2.

Idea ของ PCA

1) Select axis that preserves the maximum amount of variance
เลือกแกนที่ทำให้ข้อมูลกระจายที่สุด

2) Minimize the mean squared distance between the original dataset and its projection onto that axis
dataset ต้องอยู่ใกล้แกนให้ได้มากที่สุด

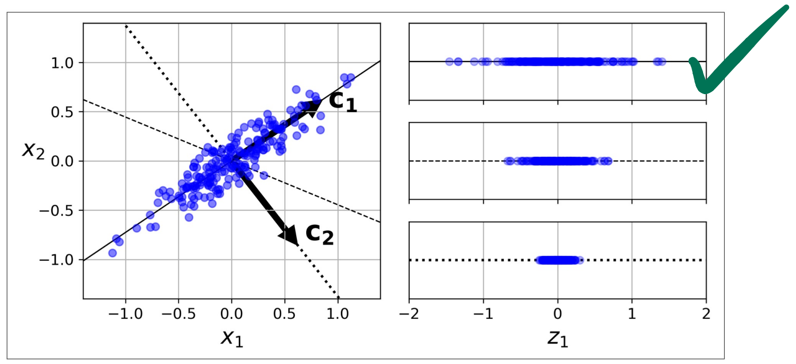


Figure 8-7. Selecting the subspace onto which to project

The unit vector that defines the i^{th} axis is called the i^{th} *principal component* (PC). In Figure 8-7, the 1^{st} PC is $\mathbf{c}_1$ and the 2^{nd} PC is $\mathbf{c}_2$. In Figure 8-2 the first two PCs are represented by the orthogonal arrows in the plane, and the third PC would be orthogonal to the plane (pointing up or down).

> The direction of the principal components is not stable: if you perturb the training set slightly and run PCA again, some of the new PCs may point in the opposite direction of the original PCs. However, they will generally still lie on the same axes. In some cases, a pair of PCs may even rotate or swap, but the plane they define will generally remain the same.

So how can you find the principal components of a training set? Luckily, there is a standard matrix factorization technique called *Singular Value Decomposition* (SVD) that can decompose the training set matrix $\mathbf{X}$ into the matrix multiplication of three matrices $\mathbf{U}\ \Sigma\ \mathbf{V}^T$, where $\mathbf{V}$ contains all the principal components that we are looking for, as shown in Equation 8-1.

Equation 8-1. Principal components matrix

$$\mathbf{V} = \begin{pmatrix} \mid & \mid & & \mid \\ \mathbf{c_1} & \mathbf{c_2} & \cdots & \mathbf{c_n} \\ \mid & \mid & & \mid \end{pmatrix}$$

Component	Meaning	Variance Captured
PC1	Direction of maximum spread	Highest
PC2	Next orthogonal direction	Second highest
PC3	Third orthogonal direction	Third highest
…	…	…

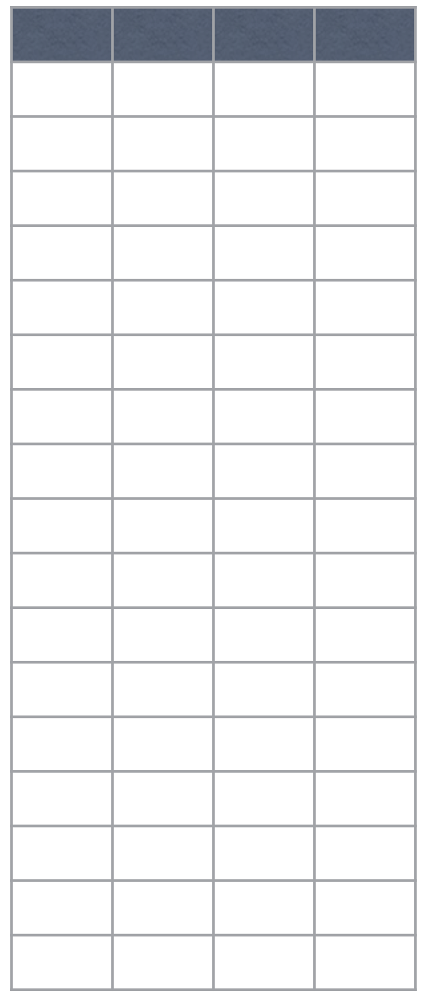
Dimensionality Reduction

Lots of variables

Row ID	Order ID	Order Date	Order Priority	Order Quantity	Sales	Discount	Ship Mode	Profit	Unit Price	Shipping Cost
1	3	10-13-10	Low	6	261.54	0.04	Regular Air	-213.25	38.94	35
49	293	10-1-12	High	49	10123.02	0.07	Delivery Truck	457.81	208.16	68.02
50	293	10-1-12	High	27	244.57	0.01	Regular Air	46.71	8.69	2.99
80	483	07-10-11	High	30	4965.7595	0.08	Regular Air	1198.97	195.99	3.99
85	515	08-28-10	Not Specified	19	394.27	0.08	Regular Air	30.94	21.78	5.94
86	515	08-28-10	Not Specified	21	146.69	0.05	Regular Air	4.43	6.64	4.95
97	613	06-17-11	High	12	93.54	0.03	Regular Air	-54.04	7.3	7.72
98	613	06-17-11	High	22	905.08	0.09	Regular Air	127.70	42.76	6.22
103	643	03-24-11	High	21	2781.82	0.07	Express Air	-695.26	138.14	35
107	678	02-26-10	Low	44	228.41	0.07	Regular Air	-226.36	4.98	8.33
127	807	11-23-10	Medium	45	196.85	0.01	Regular Air	-166.85	4.28	6.18
128	807	11-23-10	Medium	32	124.56	0.04	Regular Air	-14.33	3.95	2
134	868	06-8-12	Not Specified	32	716.84	0	Regular Air	134.72	21.78	5.94
135	868	06-8-12	Not Specified	31	1474.33	0.04	Regular Air	114.46	47.98	3.61
149	933	08-4-12	Not Specified	15	80.61	0.02	Regular Air	-4.72	5.28	2.99
160	995	05-30-11	Medium	46	1815.49	0.03	Regular Air	782.91	39.89	3.04
161	998	11-25-09	Not Specified	16	248.26	0.07	Regular Air	93.80	15.74	1.39
175	1154	02-14-12	Critical	44	4462.23	0.04	Delivery Truck	440.72	100.98	26.22
176	1154	02-14-12	Critical	11	663.784	0.25	Regular Air	-481.04	71.37	69
203	1344	04-15-12	Low	15	834.904	0.06	Regular Air	-11.68	65.99	5.26
204	1344	04-15-12	Low	18	2480.9205	0.01	Regular Air	313.58	155.99	8.99
213	1412	03-12-10	Not Specified	13	59.03	0.1	Express Air	26.92	3.69	0.5
214	1412	03-12-10	Not Specified	21	97.48	0.05	Regular Air	-5.77	4.71	0.7
229	1539	03-9-11	Low	33	511.83	0.1	Regular Air	-172.88	15.99	13.18
230	1539	03-9-11	Low	38	184.99	0.05	Regular Air	-144.55	4.89	4.93
231	1540	08-4-12	High	30	80.9	0.09	Regular Air	5.76	2.88	0.7

Principal component analysis
Manifold learning
Factor analysis
Autoencoder

Few ***transformed features***
or ***alternate representation***
that captures most input information.

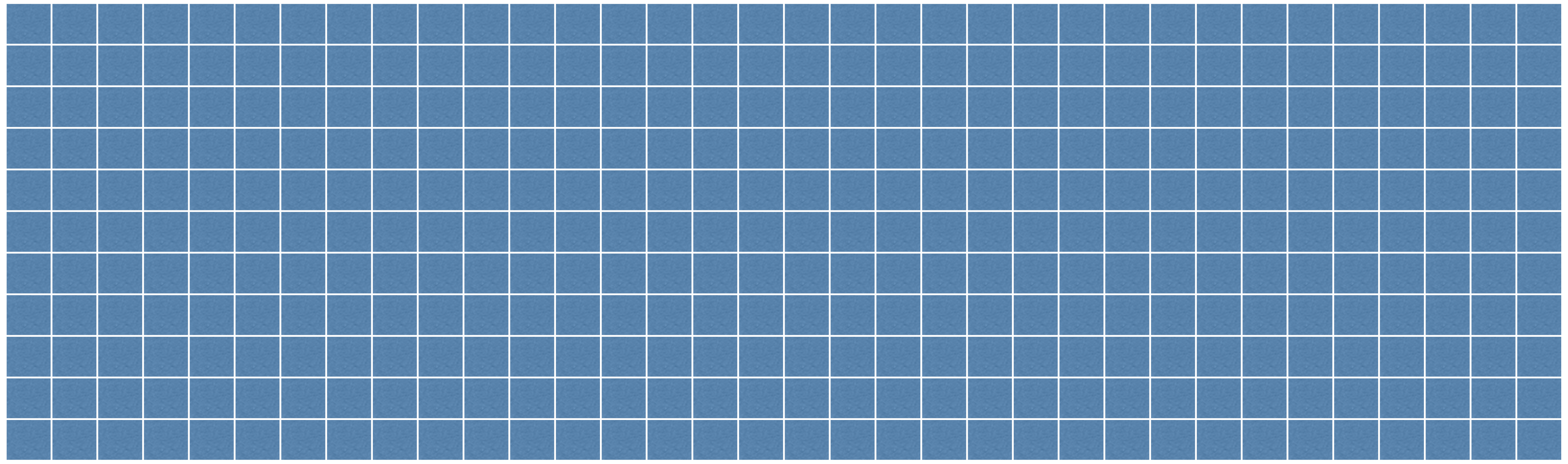


Input data

$$\boldsymbol{X}_{n\times d} = \left[\begin{array}{c|cccc} & \boldsymbol{X}_1 & \boldsymbol{X}_2 & \cdots & \boldsymbol{X}_d \\ \hline \boldsymbol{x}_1 & x_{11} & x_{12} & \cdots & x_{1d} \\ \boldsymbol{x}_2 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{x}_n & x_{n1} & x_{n2} & \cdots & x_{nd} \end{array}\right]$$

(Redundant/correlated) Original features

n rows of observations/ data points/ Users



Dimension* = *d

Notations

Row = vector of random variables

n rows of observations/ data points/ Users

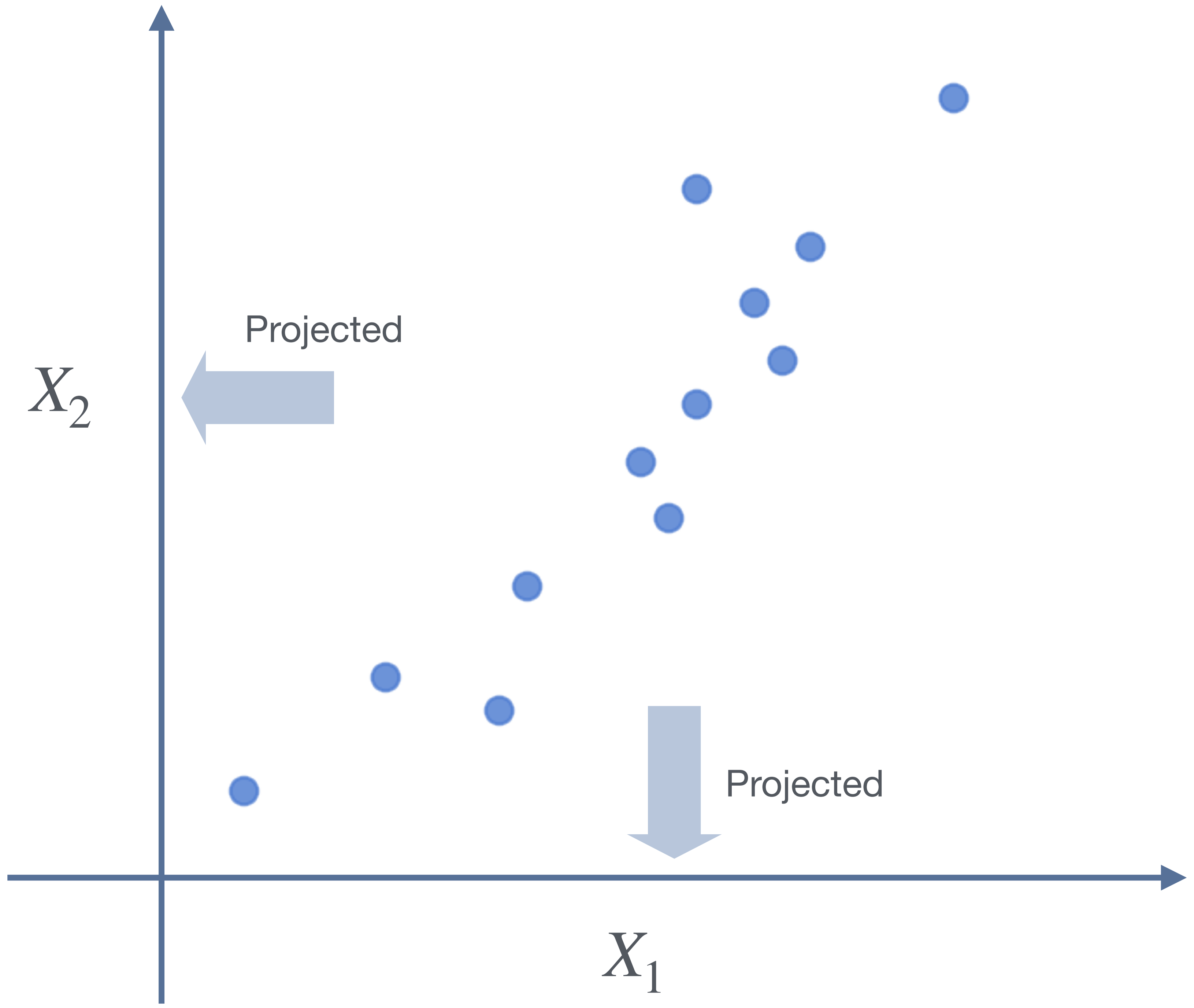
Dimension $= r \ll d$

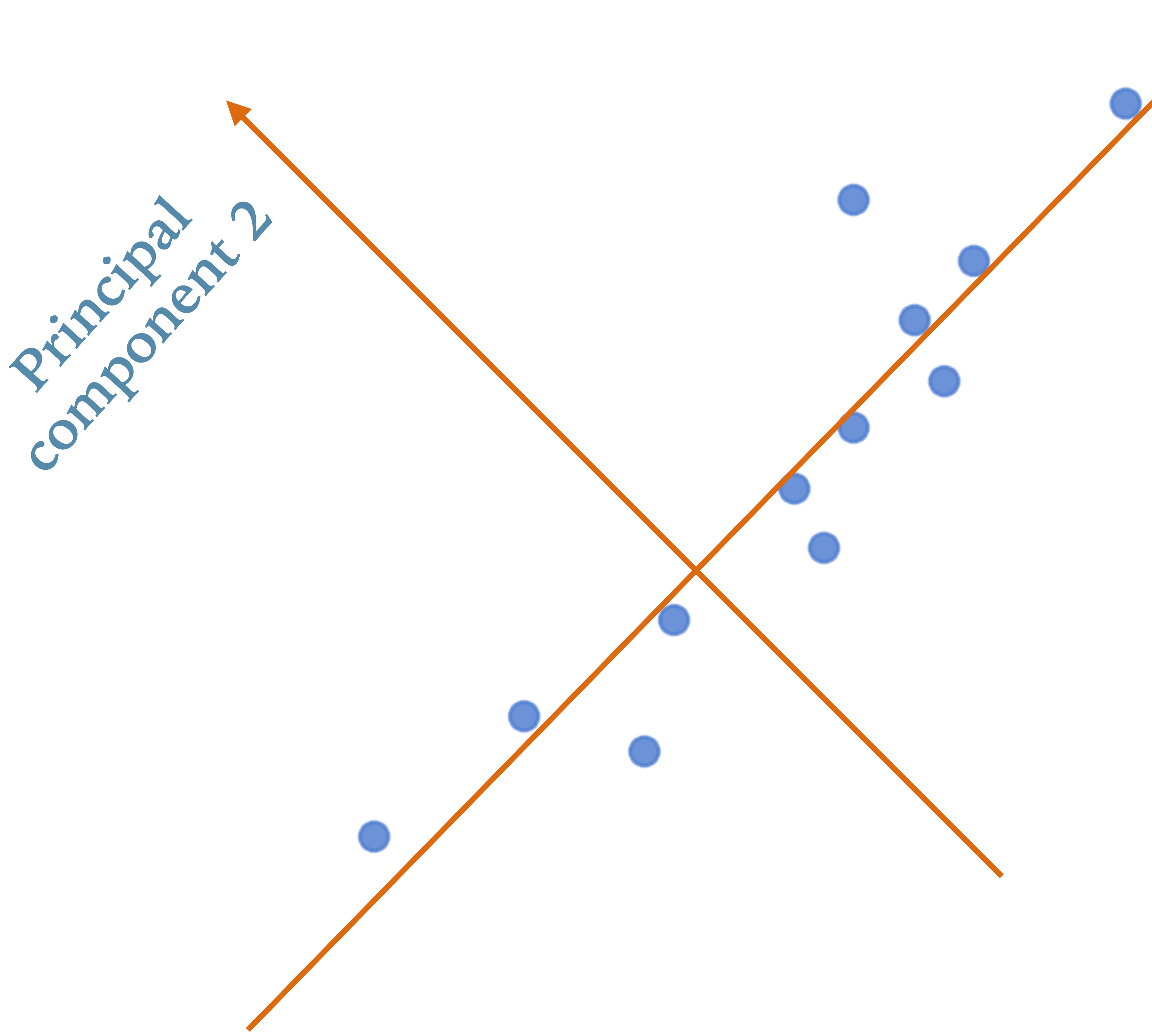
Visualization
Clustering
Supervised learning
Anomaly detection

Principal Component Analysis (PCA) : Basic Concept

```
Original data

[[ 9 39]
 [15 56]
 [25 93]
 [14 61]
 [10 50]
 [18 75]
 [ 0 32]
 [16 85]
 [ 5 42]
 [19 70]
 [16 66]
 [20 80]]
```






```
Transformed data
(PC Scores) :

[[ 23.7584  -2.8373]
 [  5.7323  -3.0802]
 [-32.5219  -0.711 ]
 [  1.3155  -0.5324]
 [ 13.0171  -0.2637]
 [-13.2283   0.1592]
 [ 33.2709   3.4487]
 [-22.0621   5.2548]
 [ 22.1966   1.9125]
 [ -8.8115  -2.3886]
 [ -4.0617  -0.8268]
 [-18.6054  -0.1352]]
```

Principal Component (PC) = Unit vector representing an axis direction.

PC 1 = Axis with PC scores having highest variance

PC 2 = Axis with PC scores having 2nd highest variance

$$\mathbf{X} \qquad \mathbf{X}-\boldsymbol{\mu} \qquad \mathbf{P} \qquad = \qquad \mathbf{Z}$$

ให้ center ที่ 0 ดูว่าต่างจาก original แค่ไหน

Original data:

```
[[ 9 39]
 [15 56]
 [25 93]
 [14 61]
 [10 50]
 [18 75]
 [ 0 32]
 [16 85]
 [ 5 42]
 [19 70]
 [16 66]
 [20 80]]
```

Zero-mean data:

```
[[ -4.92 -23.42]
 [  1.08  -6.42]
 [ 11.08  30.58]
 [  0.08  -1.42]
 [ -3.92 -12.42]
 [  4.08  12.58]
 [-13.92 -30.42]
 [  2.08  22.58]
 [ -8.92 -20.42]
 [  5.08   7.58]
 [  2.08   3.58]
 [  6.08  17.58]]
```

×

Principal components:

```
[[0.3201  0.9474]
 [0.9474 -0.3201]]
```

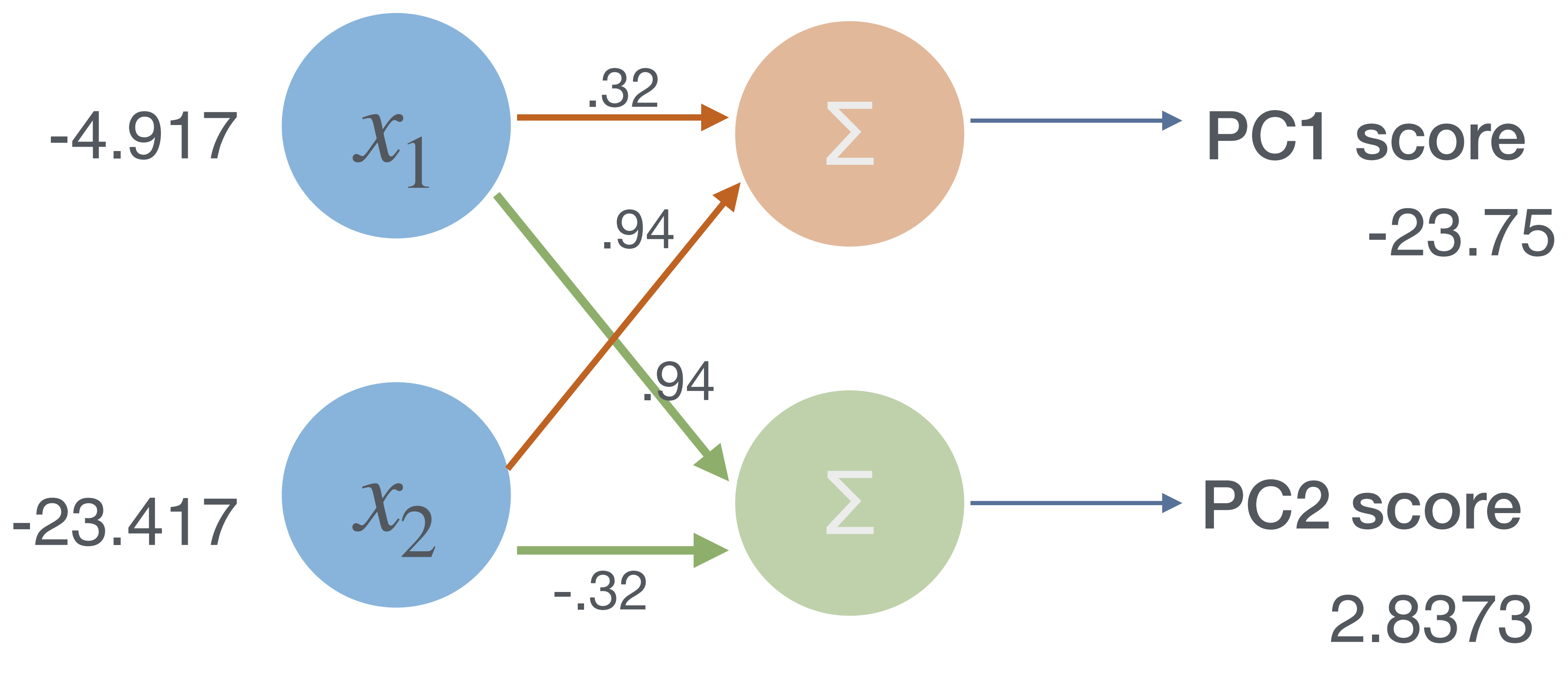
PC1	PC2
x_2 gets more weight (***loading***) in PC1	x_1 gets more weight in PC2

=

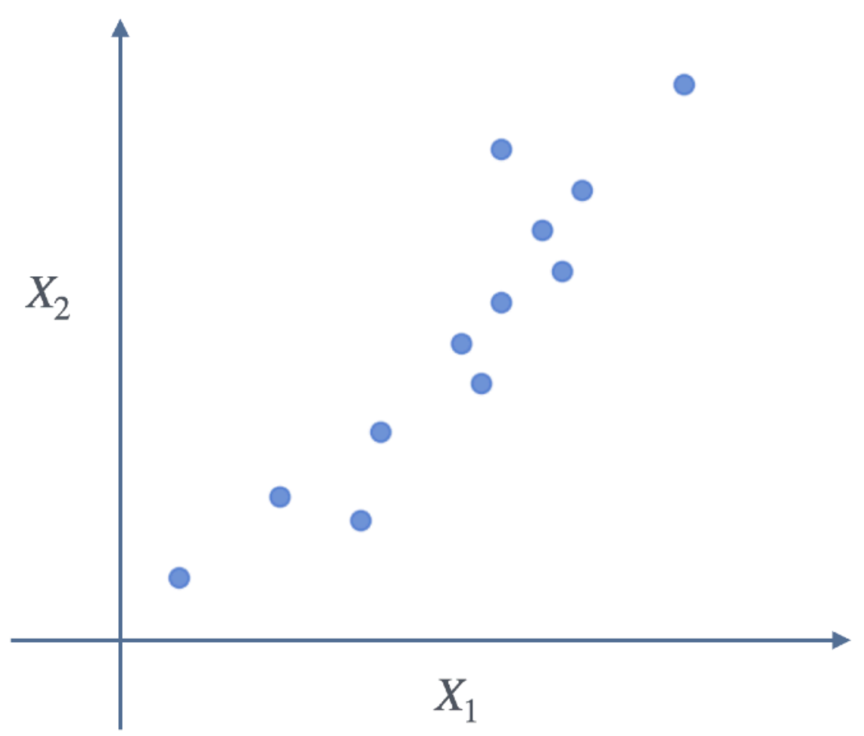
Transformed data (PC scores):

```
[[-23.7584   2.8373]
 [ -5.7323   3.0802]
 [ 32.5219   0.711 ]
 [ -1.3155   0.5324]
 [-13.0171   0.2637]
 [ 13.2283  -0.1592]
 [-33.2709  -3.4487]
 [ 22.0621  -5.2548]
 [-22.1966  -1.9125]
 [  8.8115   2.3886]
 [  4.0617   0.8268]
 [ 18.6054   0.1352]]
```

Zero column means



Proportion of Variance Explained (PVE)

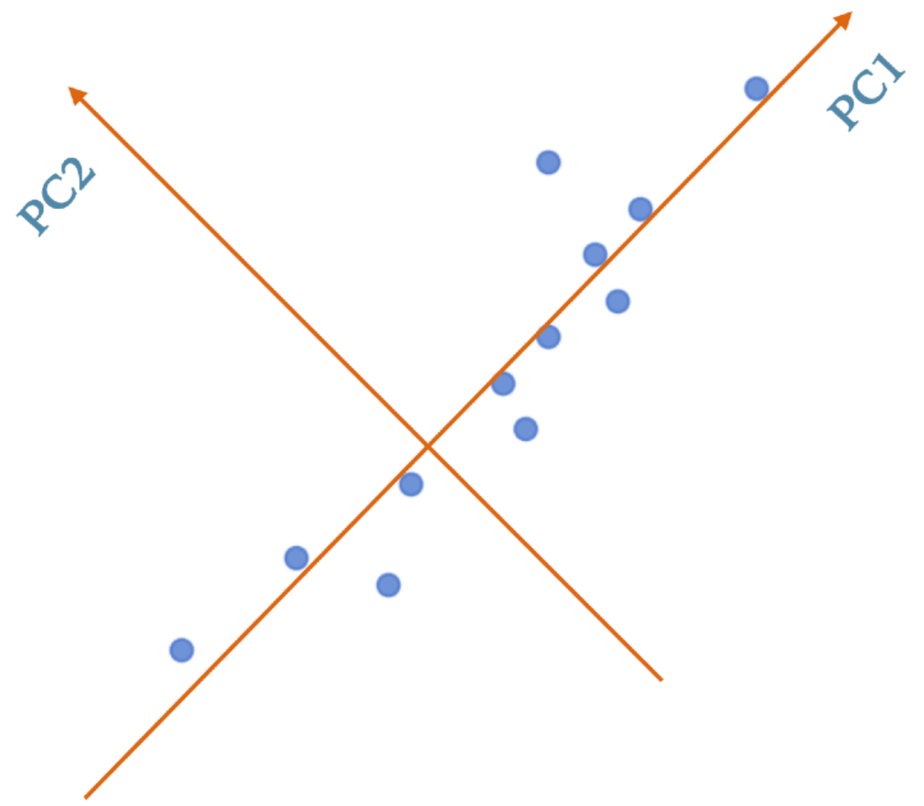



```
[[ 9 39]
 [15 56]
 [25 93]
 [14 61]
 [10 50]
 [18 75]
 [ 0 32]
 [16 85]
 [ 5 42]
 [19 70]
 [16 66]
 [20 80]]
```

Total variance = 47.7 + 370 = 417.7

All PC scores are uncorrelated and have zero means

Total variance = 411.6 + 6.18 = 417.7



PC1 scores   PC2 scores

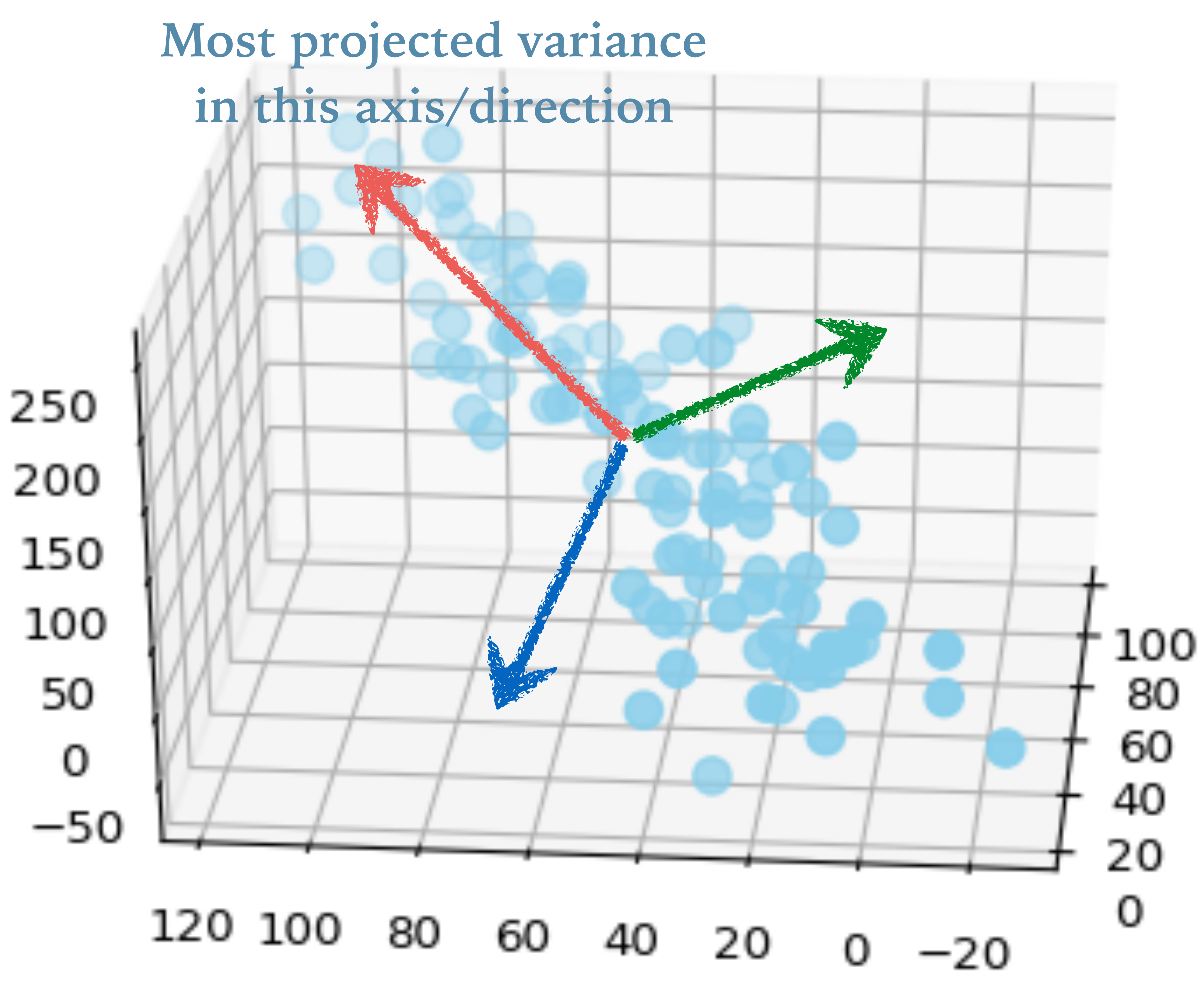
```
[[ 23.7584  -2.8373]
 [  5.7323  -3.0802]
 [-32.5219  -0.711 ]
 [  1.3155  -0.5324]
 [ 13.0171  -0.2637]
 [-13.2283   0.1592]
 [ 33.2709   3.4487]
 [-22.0621   5.2548]
 [ 22.1966   1.9125]
 [ -8.8115  -2.3886]
 [ -4.0617  -0.8268]
 [-18.6054  -0.1352]]
```

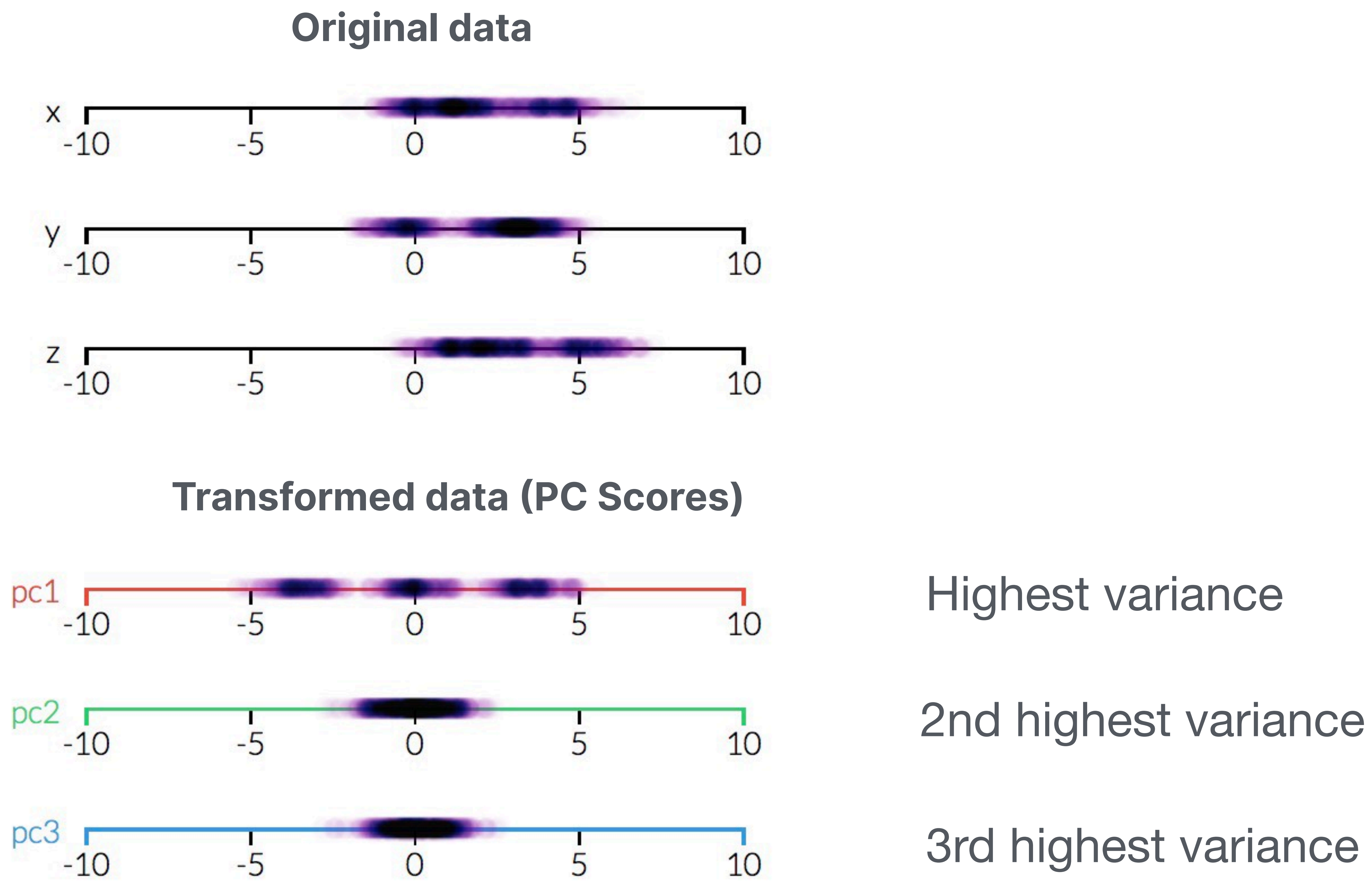
PVE by PC1 ~ 98.52%

PVE by PC2 ~ 1.47%

Total variance of original data = Total variance of **all PC scores**.

Since PC1 scores capture most information in the original data, the dimension thus can be reduced from 2 to 1.

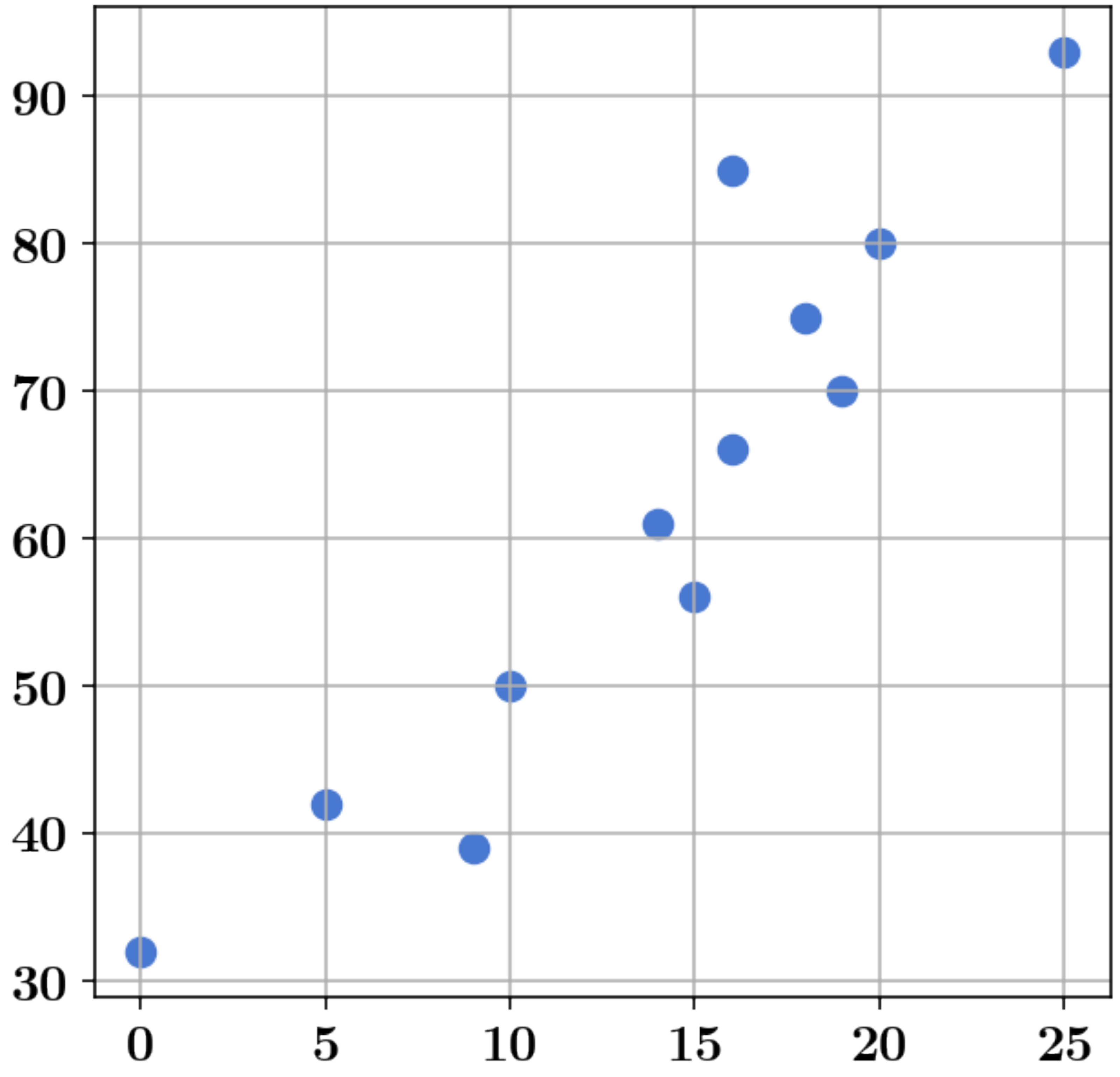




Data dimension could be reduced from 3 to 2.

Sample Covariance of Two Centered Variables

	Raw data before centering		$\mathbf{X}_1$	$\mathbf{X}_2$	$\mathbf{X}_1 \cdot \mathbf{X}_2$
$\mathbf{x}_1$	9	39	-4.92	-23.42	115.23
$\mathbf{x}_2$	15	56	1.08	-6.42	-6.93
⋮	25	93	11.08	30.58	338.83
	14	61	0.08	-1.42	-0.11
	10	50	-3.92	-12.42	48.69
	18	75	4.08	12.58	51.33
	0	32	-13.92	-30.42	423.45
	16	85	2.08	22.58	46.97
	5	42	-8.92	-20.42	182.15
	19	70	5.08	7.58	38.51
	16	66	2.08	3.58	7.45
$\mathbf{x}_n$	20	80	6.08	17.58	106.89



$$\text{(Sample) } \mathbb{C}\text{ov}(X_1, X_2) = \frac{\sum_{i=1}^{n}(x_{i1} - \overline{X}_1)(x_{i2} - \overline{X}_2)}{n-1}$$

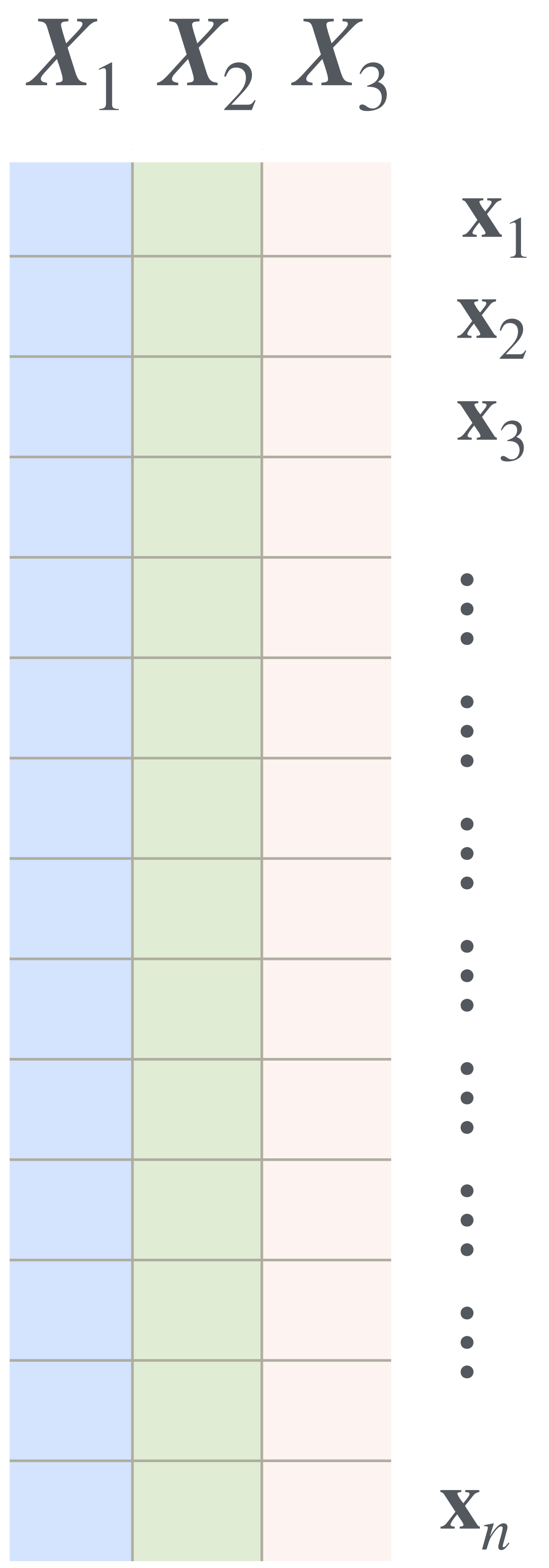
$$= \frac{\boldsymbol{X}_1 \cdot \boldsymbol{X}_2}{n-1} \quad \text{if } \boldsymbol{X}_1 \text{ and } \boldsymbol{X}_2 \text{ have zero mean}$$

$$\mathbb{C}\text{ov}(X_1, X_1) = \mathbb{V}\text{ar}(X_1)$$

Sample Covariance Matrix of a Random Vector

- Represent all pair-wise linear relationships in dataset.
- Let $\mathbf{X}$ be an $(n \times d)$ data matrix with ***zero-mean*** columns $\{\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_d\}$ and rows $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$

$\boldsymbol{X} =$



$$\boldsymbol{\Sigma} = \begin{bmatrix} \mathbb{C}\mathrm{ov}(X_1, X_1) & \mathbb{C}\mathrm{ov}(X_1, X_2) & \mathbb{C}\mathrm{ov}(X_1, X_3) \\ \mathbb{C}\mathrm{ov}(X_2, X_1) & \mathbb{C}\mathrm{ov}(X_2, X_2) & \mathbb{C}\mathrm{ov}(X_2, X_2) \\ \mathbb{C}\mathrm{ov}(X_3, X_1) & \mathbb{C}\mathrm{ov}(X_3, X_2) & \mathbb{C}\mathrm{ov}(X_3, X_3) \end{bmatrix}$$

$$= \frac{1}{n-1} \begin{bmatrix} \boldsymbol{X}_1 \cdot \boldsymbol{X}_1 & \boldsymbol{X}_1 \cdot \boldsymbol{X}_2 & \boldsymbol{X}_1 \cdot \boldsymbol{X}_3 \\ \boldsymbol{X}_2 \cdot \boldsymbol{X}_1 & \boldsymbol{X}_2 \cdot \boldsymbol{X}_2 & \boldsymbol{X}_2 \cdot \boldsymbol{X}_3 \\ \boldsymbol{X}_3 \cdot \boldsymbol{X}_1 & \boldsymbol{X}_3 \cdot \boldsymbol{X}_2 & \boldsymbol{X}_3 \cdot \boldsymbol{X}_3 \end{bmatrix}$$

$$= \frac{1}{n-1} \boldsymbol{X}^T \boldsymbol{X}$$

$$= \frac{1}{n-1} \sum_{i=1}^{n} \boldsymbol{x}_i^T \boldsymbol{x}_i = \frac{1}{n-1} \sum_{i=1}^{n} \begin{bmatrix} x_{i1}^2 & x_{i1}x_{i2} & \cdots & x_{i1}x_{id} \\ x_{i2}x_{i1} & x_{i2}^2 & \cdots & x_{i2}x_{i3} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i3}x_{i1} & x_{i3}x_{i2} & \cdots & x_{id}^2 \end{bmatrix}$$

	$\mathbf{X}_1$	$\mathbf{X}_2$	$\mathbf{X}_1 \cdot \mathbf{X}_2$
	-4.92	-23.42	115.23
	1.08	-6.42	-6.93
	11.08	30.58	338.83
	0.08	-1.42	-0.11
	-3.92	-12.42	48.69
	4.08	12.58	51.33
	-13.92	-30.42	423.45
	2.08	22.58	46.97
	-8.92	-20.42	182.15
	5.08	7.58	38.51
	2.08	3.58	7.45
	6.08	17.58	106.89
(Squared Total)	954	7400	1352.4
Average	47.7	370	122.9

$$\mathbf{\Sigma} \approx \begin{bmatrix} 47.7 & 122.9 \\ 122.9 & 370 \end{bmatrix}$$

Variance of each variable = $\text{var}(X_i) = \sum_j (X_{ij} - \overline{X}_i)^2/(n-1)$

Total variance in the dataset = $\sum_i \text{var}(X_i) = 417.7$

Given the sample covariance matrix $\boldsymbol{\Sigma}$ of the input data, project the data on the new axes (PCs) so that

$$\boldsymbol{\Sigma} \quad \Rightarrow \quad \boldsymbol{\Sigma}_Z$$

Original covariance matrix

$$\begin{bmatrix} 47.7 & 122.9 \\ 122.9 & 370 \end{bmatrix}$$

$$\begin{bmatrix} 411.62 & 0 \\ 0 & 6.18 \end{bmatrix}$$

New covariance matrix with only diagonal entries (All PCs uncorrelated)

☐ Let $\mathbf{P} = [\mathbf{p}_1\ \mathbf{p}_2\ \ \dots\ \ \mathbf{p}_d]$ and $\{\lambda_1, \lambda_2, \dots, \lambda_d\}$ respectively be eigenvectors and eigenvalues of $\mathbf{\Sigma}$ (largest ones first).

$$\mathbf{\Sigma} = \begin{bmatrix} 47.7 & 122.9 \\ 122.9 & 370 \end{bmatrix}$$

$$\mathbf{\Sigma}\, \boldsymbol{p}_1 = \lambda_1 \boldsymbol{p}_1$$

$$\mathbf{\Sigma}\, \boldsymbol{p}_2 = \lambda_2 \boldsymbol{p}_2$$

$$\boldsymbol{P} = \begin{bmatrix} \boldsymbol{p}_1 & \boldsymbol{p}_2 \end{bmatrix} = \begin{bmatrix} -0.32 & -0.947 \\ -0.947 & 0.32 \end{bmatrix}$$

Eigenvector($\mathbf{\Sigma}$) with largest eigenvalues first

$\lambda_1 = 411.6$ $\quad$ $\lambda_1 = 6.18$

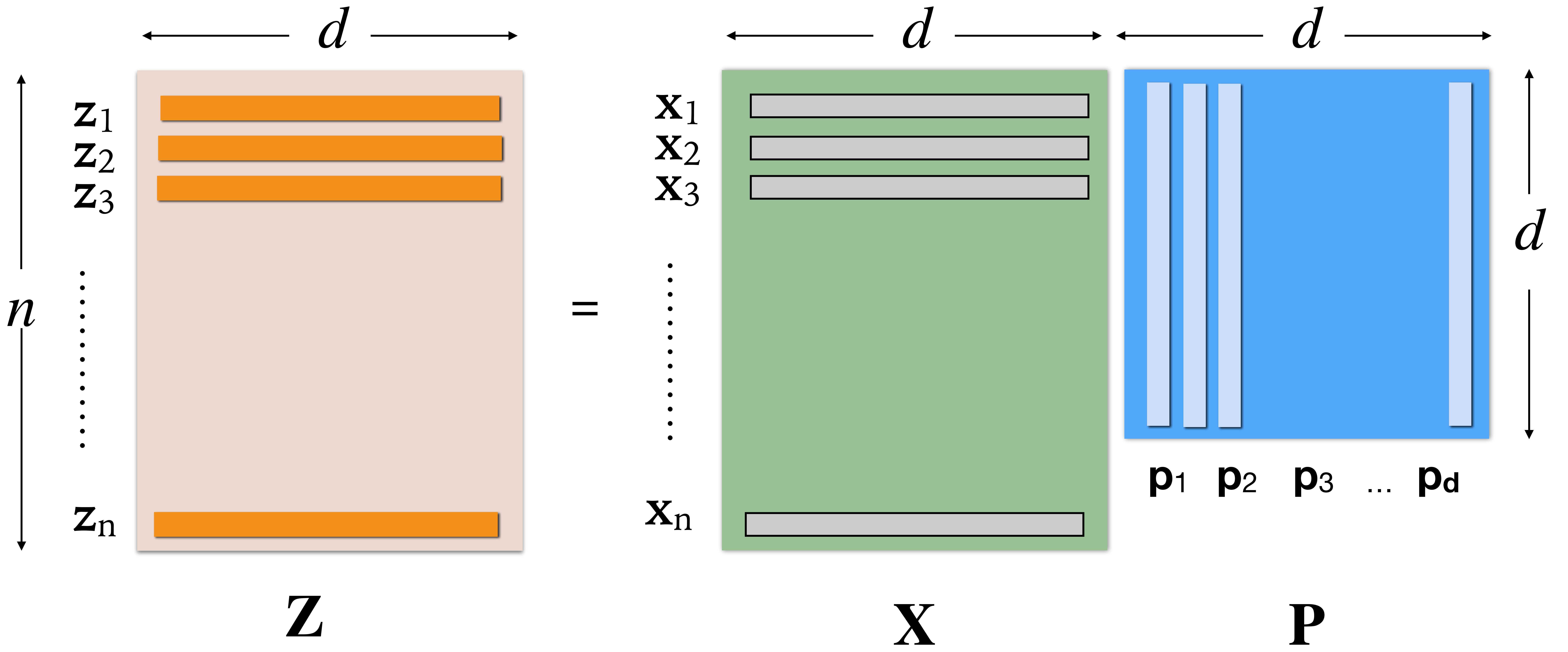
☐ Then, the following transformation on zero-mean $\mathbf{X}$ results in the data with a diagonal covariance matrix:

$$\boldsymbol{Z} = \boldsymbol{X}\boldsymbol{P}$$

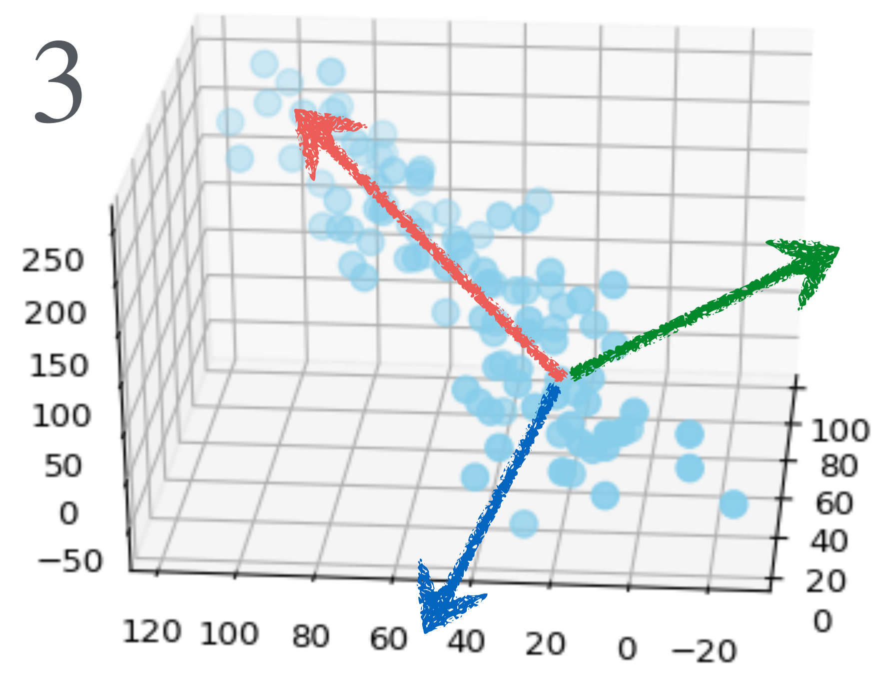
```
[[ -4.917 -23.417]
 [  1.083  -6.417]
 [ 11.083  30.583]
 [  0.083  -1.417]
 [ -3.917 -12.417]
 [  4.083  12.583]
 [-13.917 -30.417]
 [  2.083  22.583]
 [ -8.917 -20.417]
 [  5.083   7.583]
 [  2.083   3.583]
 [  6.083  17.583]]
```

```
[[ -0.320 -0.947]
 [ -0.947  0.320]]
```

Transformed data



$d = 3$

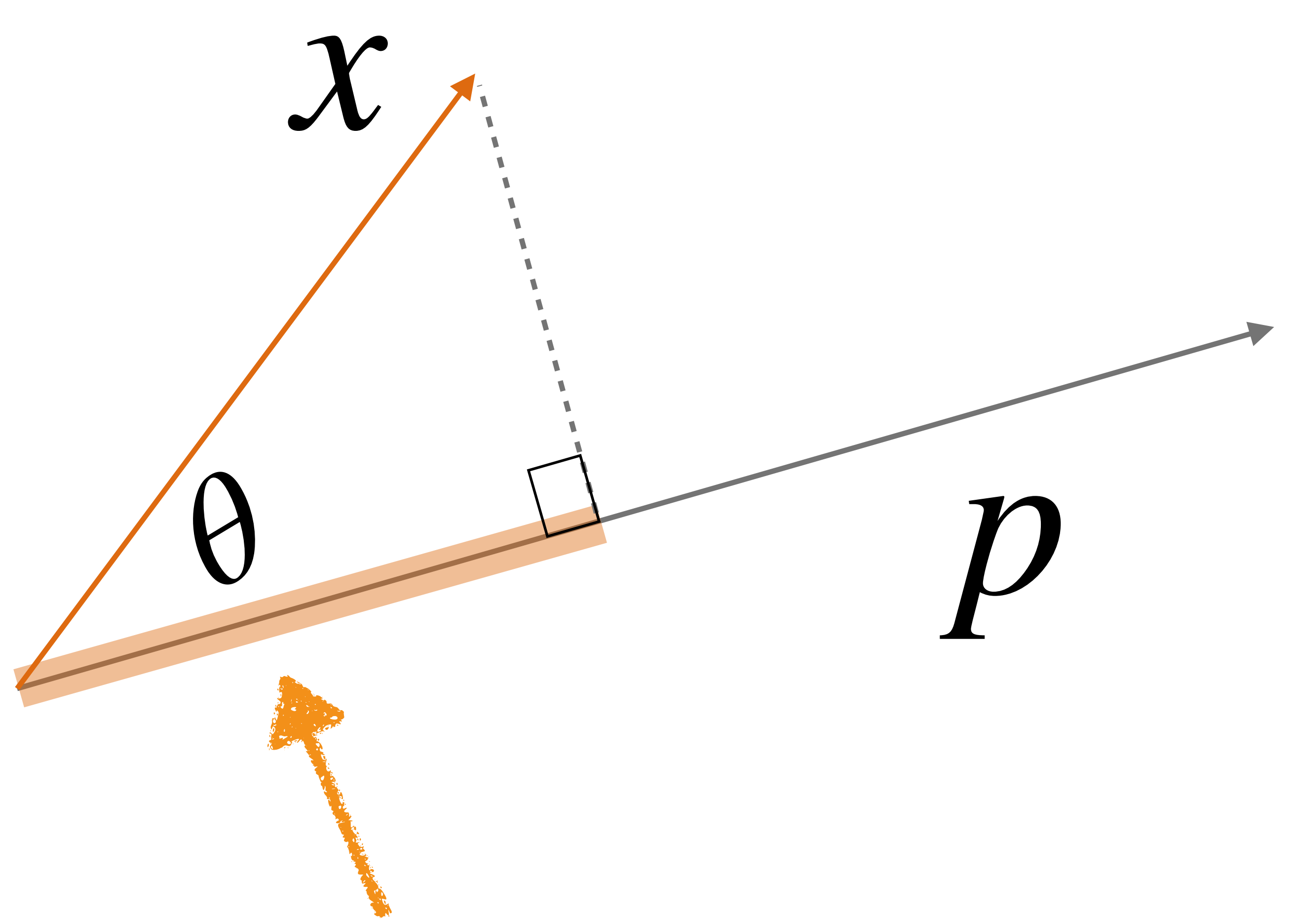


$$\mathbf{z}_i = \mathbf{x}_i \mathbf{P}$$

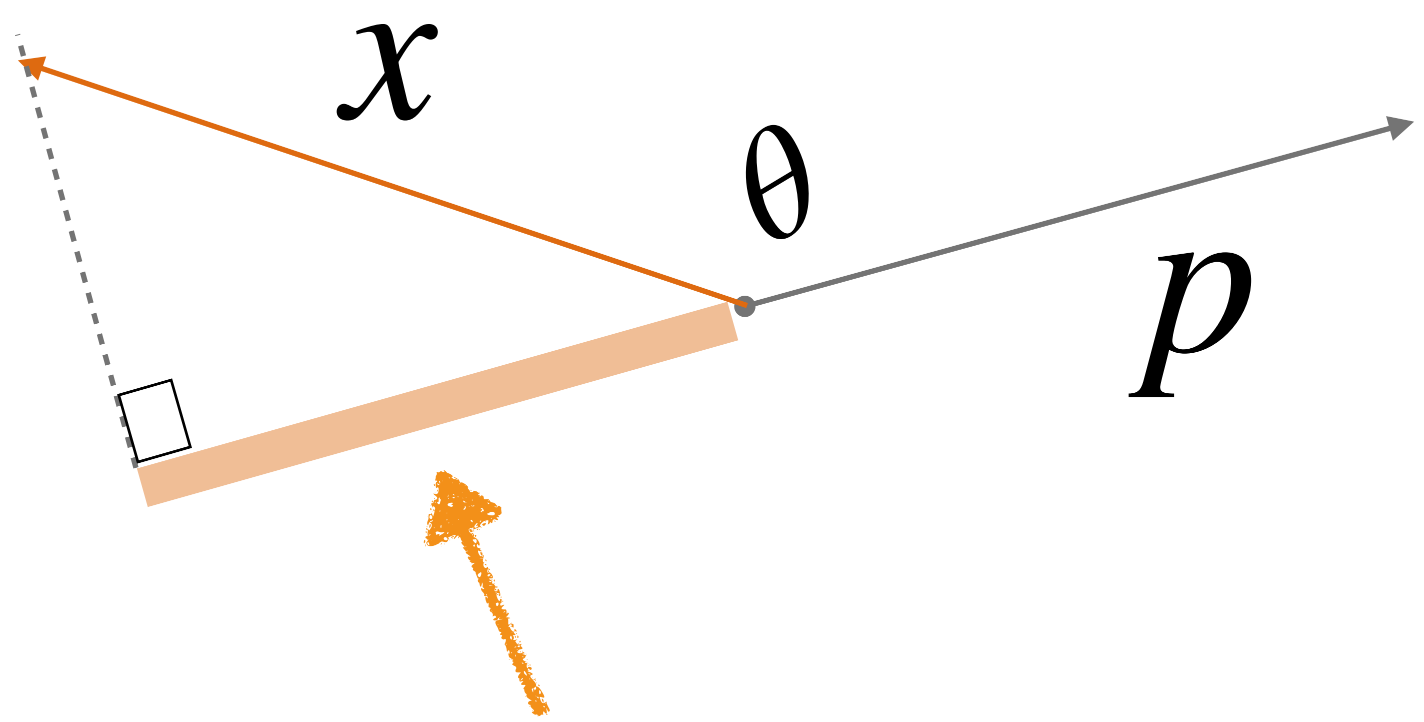
$$= [\mathbf{x}_i \cdot \mathbf{p}_1 \;\; \mathbf{x}_i \cdot \mathbf{p}_2 \;\; \cdots \;\; \mathbf{x}_i \cdot \mathbf{p}_d]$$

Scalar Projection

- Project "size" of one vector on another vector direction.

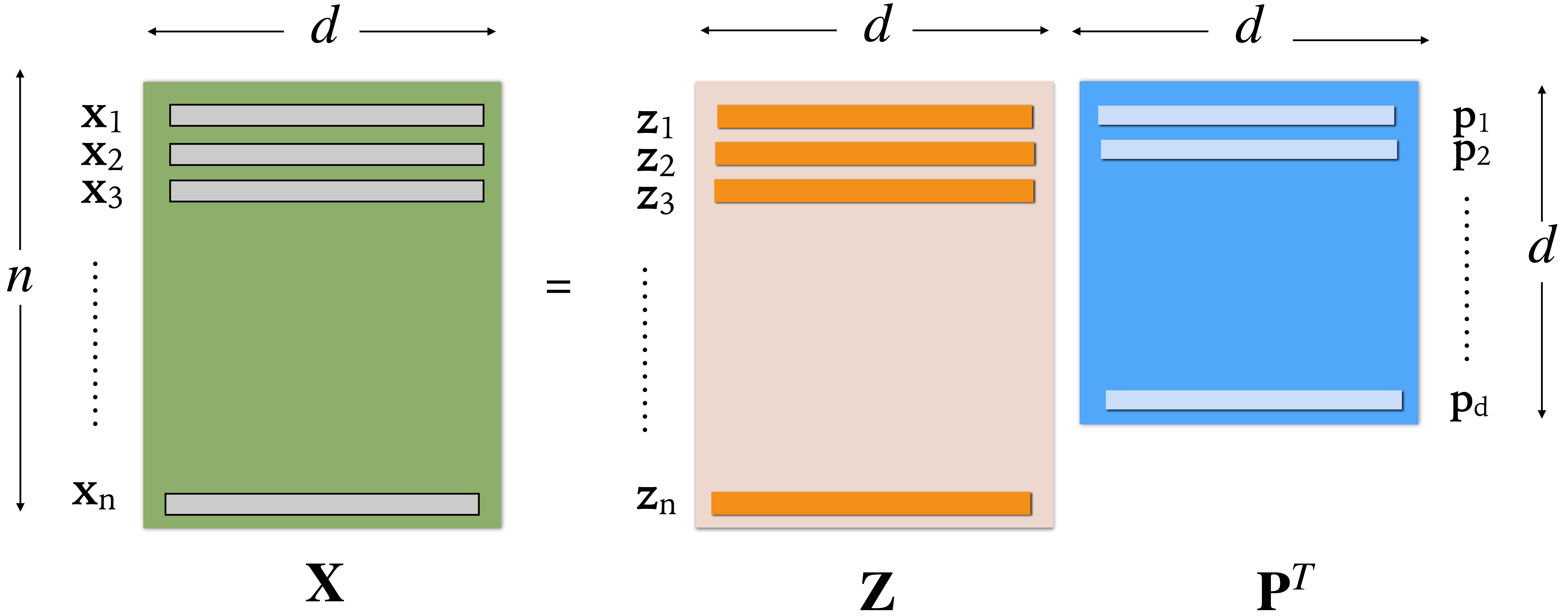


$$\mathrm{Proj}_{\boldsymbol{p}}\boldsymbol{x} = \boldsymbol{x} \cdot \frac{\boldsymbol{p}}{\|\boldsymbol{p}\|}$$



$$\mathrm{Proj}_{\boldsymbol{p}}\boldsymbol{x} = \boldsymbol{x} \cdot \frac{\boldsymbol{p}}{\|\boldsymbol{p}\|}$$

Inverse transformed data



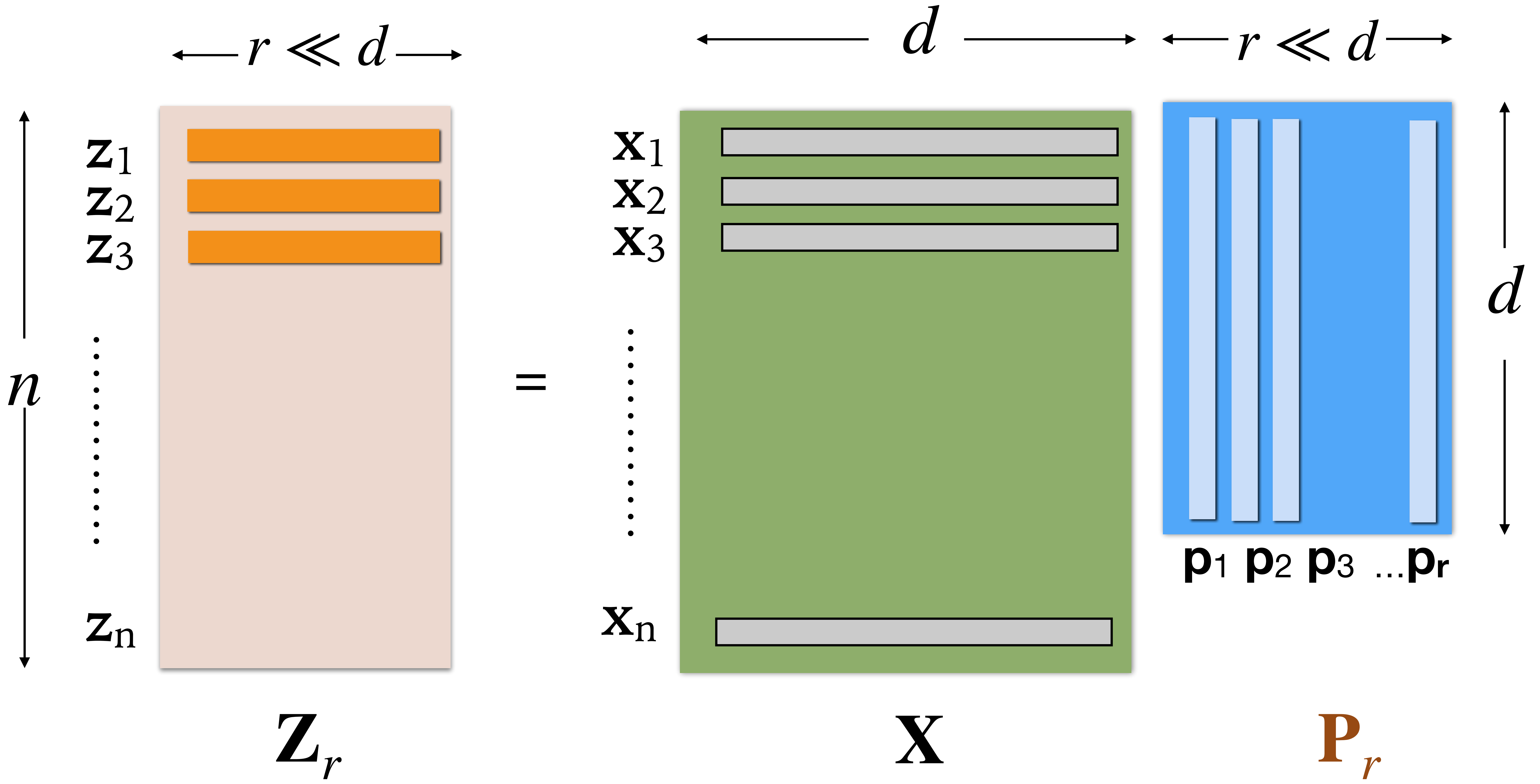
$$\boldsymbol{X}_{n\times d} = \begin{array}{c|cccc} & \boldsymbol{X}_1 & \boldsymbol{X}_2 & \cdots & \boldsymbol{X}_d \\ \hline \boldsymbol{x}_1 & x_{11} & x_{12} & \cdots & x_{1d} \\ \boldsymbol{x}_2 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{x}_n & x_{n1} & x_{n2} & \cdots & x_{nd} \end{array}$$

$$\mathbf{z}_i = [z_{i1}, z_{i2}, \cdots, z_{id}]$$

$$\mathbf{x}_i = z_{i1}\mathbf{p}_1^T + z_{i2}\mathbf{p}_2^T + \cdots + z_{id}\mathbf{p}_d^T$$

$$\boldsymbol{X} = \boldsymbol{Z}\,\boldsymbol{P}^T \quad (+\text{subtracted mean})$$

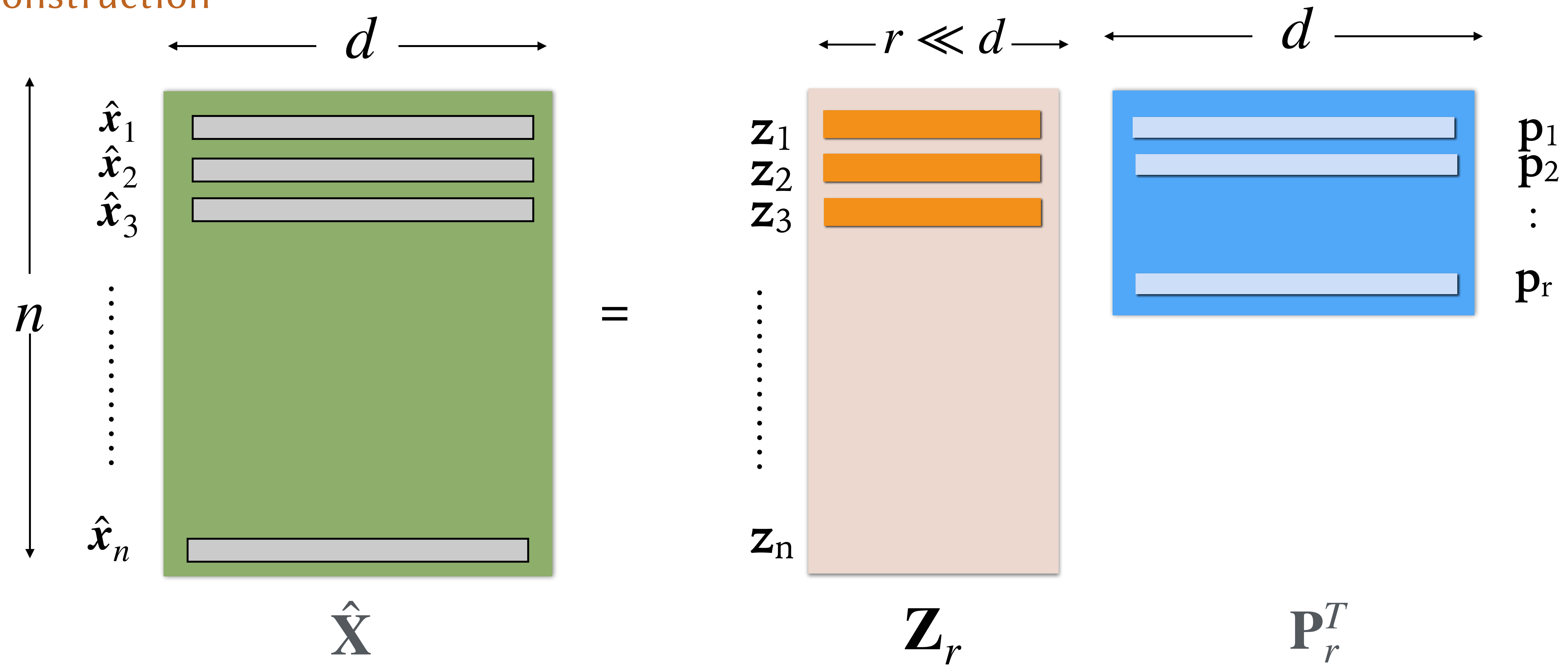
Reducing Dimension



$\mathbf{z}_i = \mathbf{x}_i \mathbf{P}_r$

Entries = First r Eigenvectors of Σ
with largest eigenvalues first

Reconstruction



$$\hat{X}_{n\times d} = \begin{array}{c|cccc} & \boldsymbol{X}_1 & \boldsymbol{X}_2 & \cdots & \boldsymbol{X}_d \\ \hline \hat{\boldsymbol{x}}_1 & x_{11} & x_{12} & \cdots & x_{1d} \\ \hat{\boldsymbol{x}}_2 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \hat{\boldsymbol{x}}_n & x_{n1} & x_{n2} & \cdots & x_{nd} \end{array}$$

$$\mathbf{z}_i = [z_{i1}, z_{i2}, \cdots, z_{ir}]$$

$$\hat{\mathbf{x}}_i = z_{i1}\mathbf{p}_1^T + z_{i2}\mathbf{p}_2^T + \cdots + z_{ir}\mathbf{p}_r^T$$

$$\hat{\boldsymbol{X}} = \boldsymbol{Z}_r \boldsymbol{P}_r^T \quad (+\text{subtracted mean})$$

Variance Properties of Projected Data Set in Reduced Dimension

$$\boldsymbol{\Sigma}_Z = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & \lambda_r \end{bmatrix}$$

$\lambda_i =$ Eigval($\boldsymbol{\Sigma}$) = Variance of PC i

Total variance of original data $= \lambda_1 + \lambda_2 + \cdots + \lambda_d$

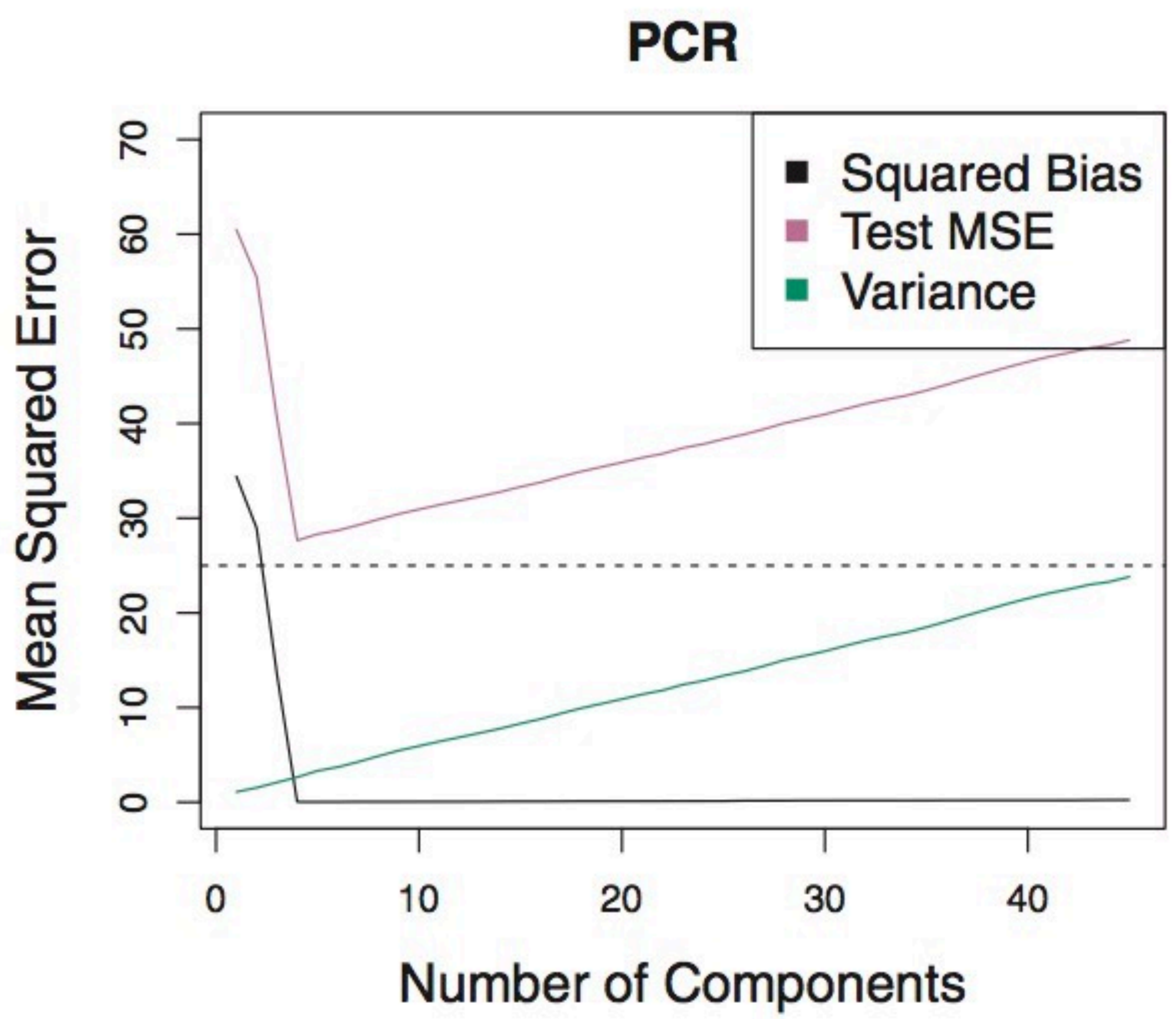
Total variance of PC scores $= \lambda_1 + \lambda_2 + \cdots + \lambda_r$

Proportion of variance explained (PVE) by first r PCs:

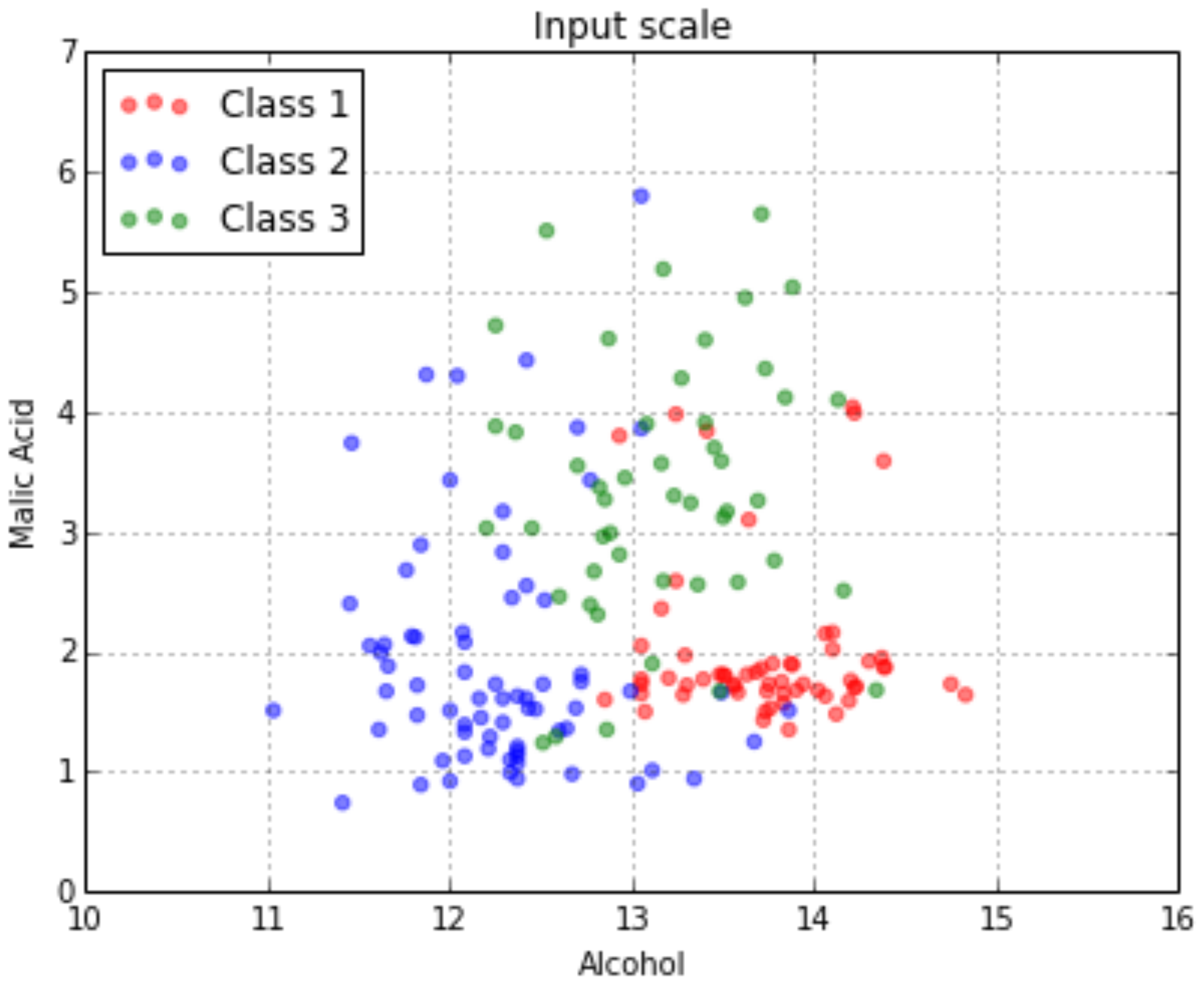
$$f(r) = \frac{\lambda_1 + \lambda_2 + \cdots + \lambda_r}{\lambda_1 + \lambda_2 + \cdots + \lambda_d} = \frac{\sum_{i=1}^{r} \lambda_i}{\sum_{i=1}^{d} \lambda_i}$$

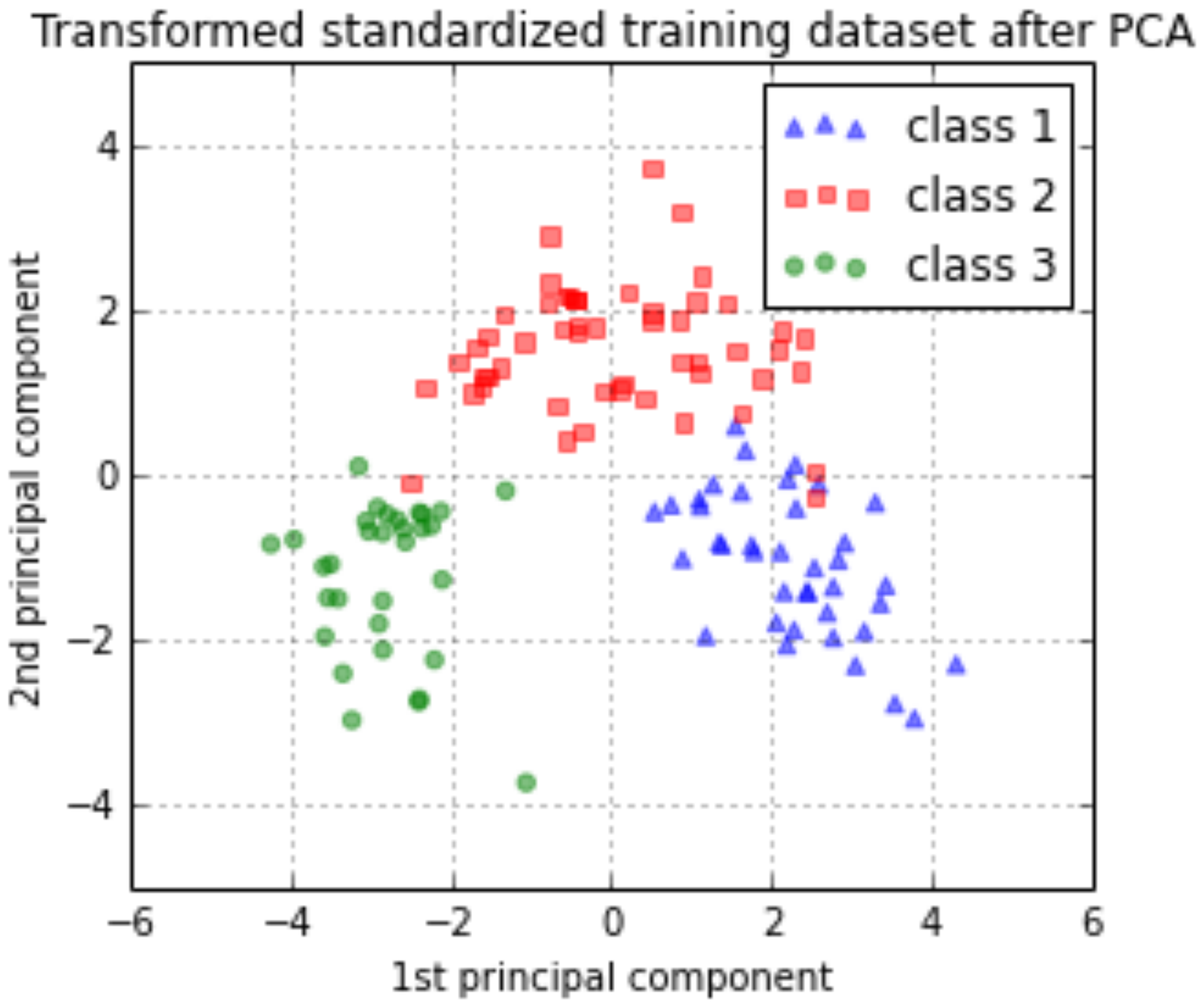
$$\begin{bmatrix} 411.62 & 0 \\ 0 & 6.18 \end{bmatrix}$$

Applying PCA to Supervised Learning



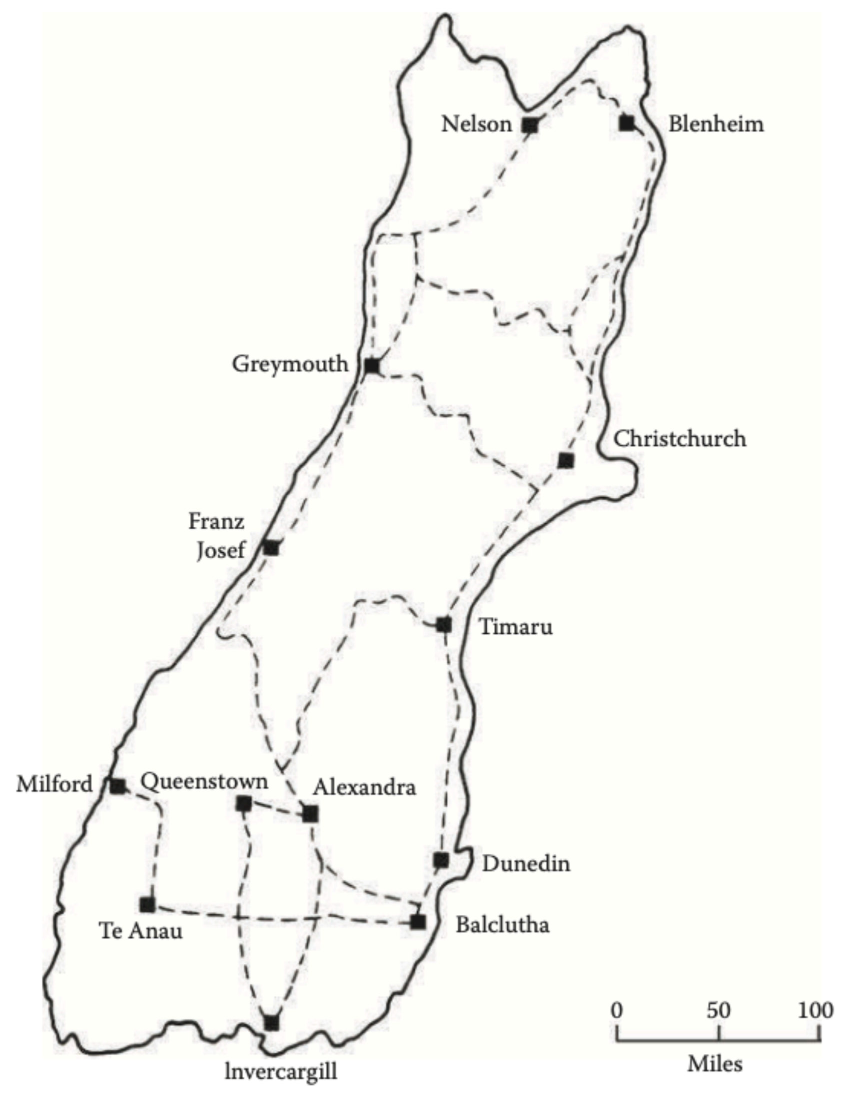
Principal component regression (PCR)





Could yield bettter class separability and guide the choices of classification algorithm.

- ☐ Given a map, can you generate some intercity distances ?
- ☐ Given a set of distances, can you recreate the (2D) map itself ?



	Berlin	Dresden	Hamburg	Koblenz	Munich	Rostock
Berlin	0	214	279	610	596	237
Dresden		0	492	533	496	444
Hamburg			0	520	772	140
Koblenz				0	521	687
Munich					0	771
Rostock						0

- ☐ Applications in political science, psychology, marketing, etc.

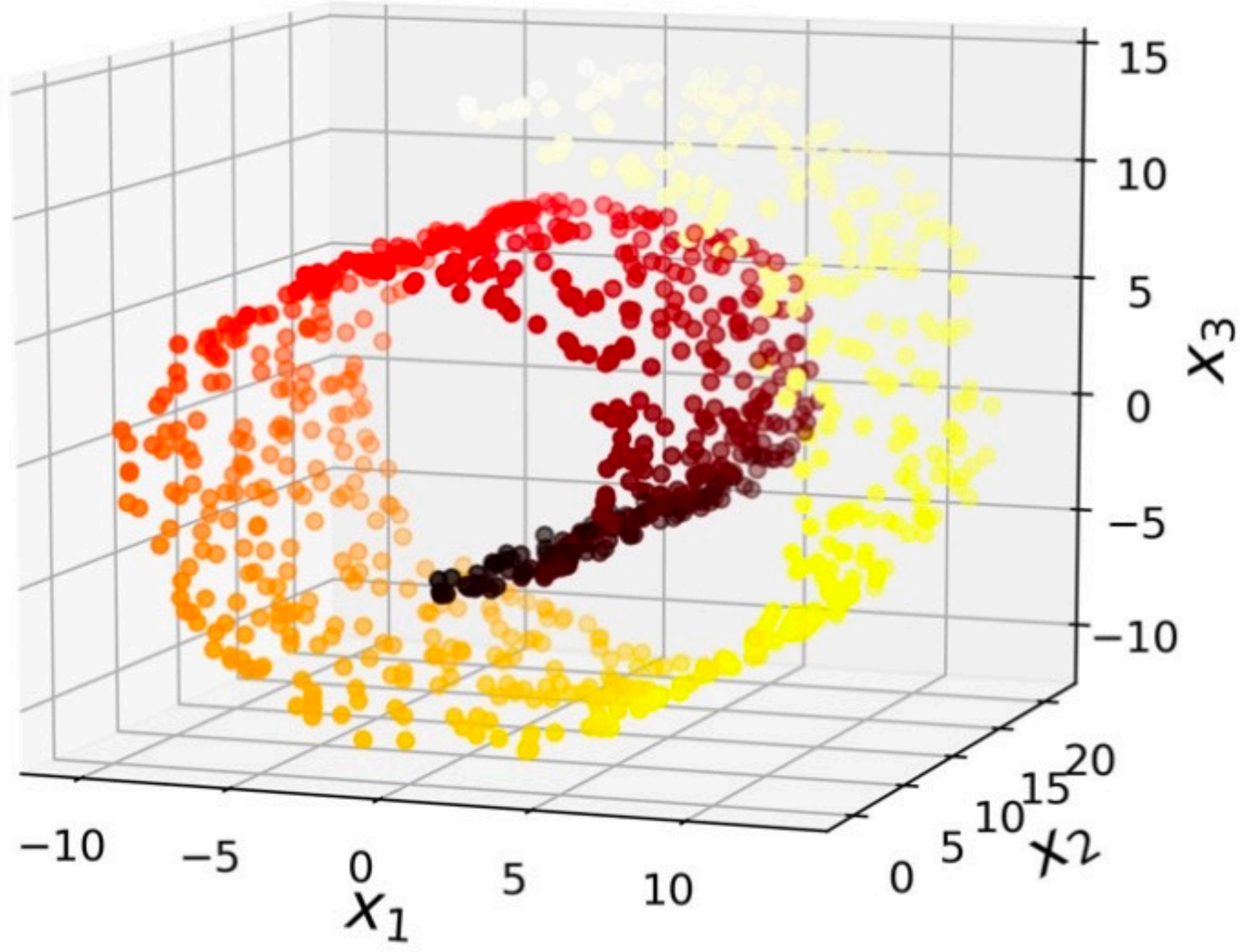
Table 11.5 The distances between 15 congressmen from New Jersey in the United States House of Representatives

	Hunt	Sandman	Howard	Thompson	Frelinghuysen	Forsythe	Widnall	Roe	Helstoski	Rodino	Minish	Rinaldo	Maraziti	Daniels	Pattern
Hunt (R)	0														
Sandman (R)	8	0													
Howard (D)	15	17	0												
Thompson (D)	15	12	9	0											
Frelinghuysen (R)	10	13	16	14	0										
Forsythe (R)	9	13	12	12	8	0									
Widnall (R)	7	12	15	13	9	7	0								
Roe (D)	15	16	5	10	13	12	17	0							
Helstoski (D)	16	17	5	8	14	11	16	4	0						
Rodino (D)	14	15	6	8	12	10	15	5	3	0					
Minish (D)	15	16	5	8	12	9	14	5	2	1	0				
Rinaldo (R)	16	17	4	6	12	10	15	3	1	2	1	0			
Maraziti (R)	7	13	11	15	10	6	10	12	13	11	12	12	0		
Daniels (D)	11	12	10	10	11	6	11	7	7	4	5	6	9	0	
Pattern (D)	13	16	7	7	11	10	13	6	5	6	5	4	13	9	0

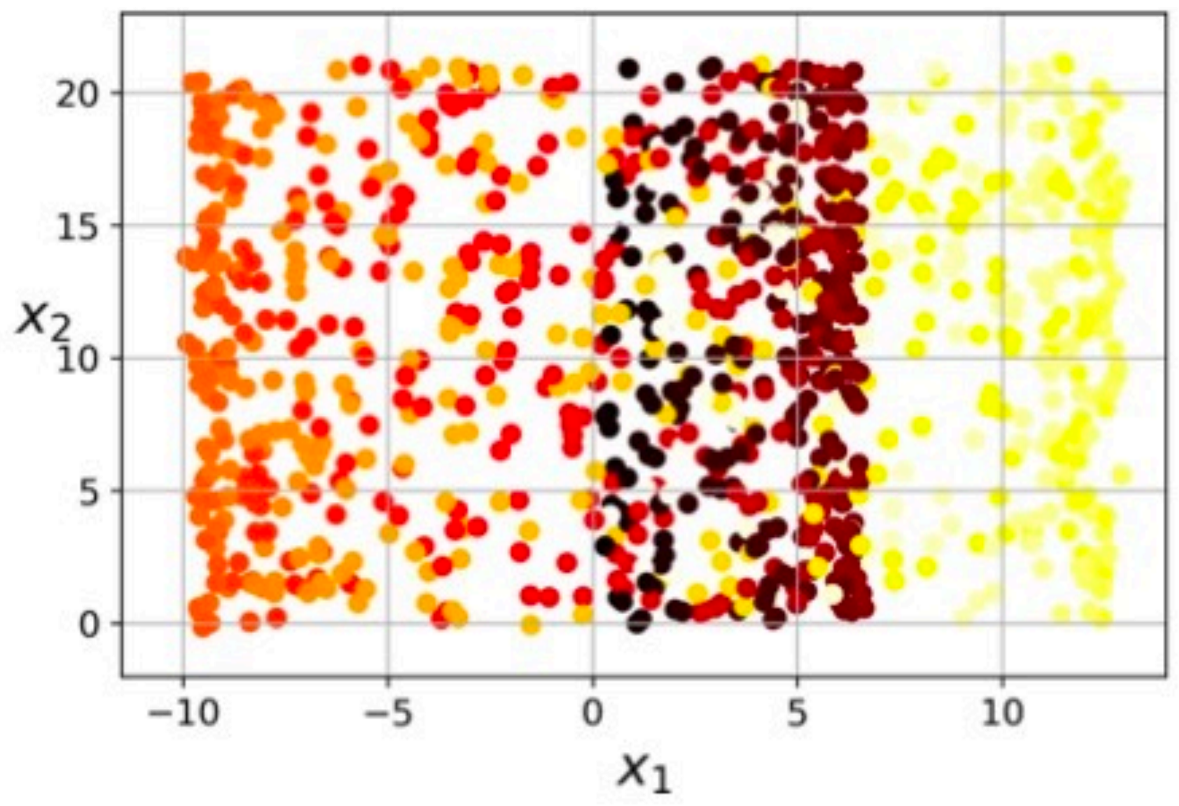
Note: The numbers shown are the number of times that the congressmen voted differently on 19 environmental bills. R = Republican party, D = Democratic party.

Manifold Learning

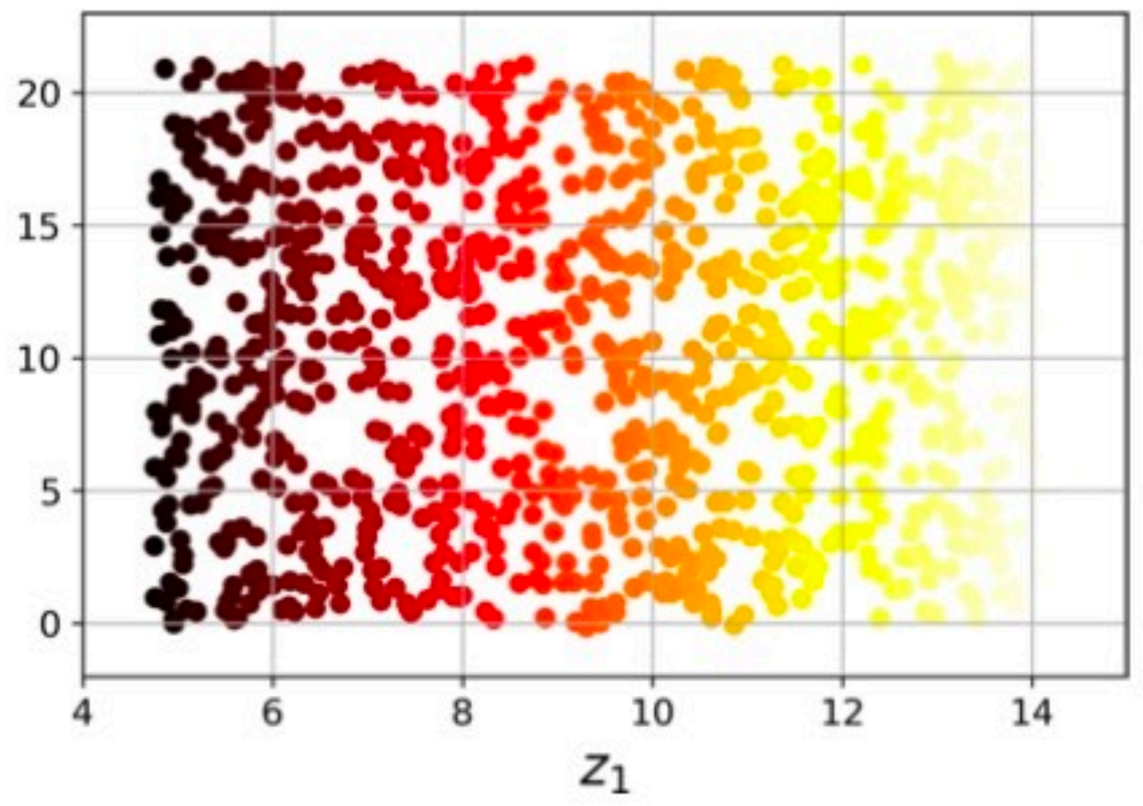
- Discover low-dimensional structure (manifold) in R^q on which the data actually lies from a higher dimensional input space R^d.
- Learning types are either ***linear*** or ***non-linear***.



Original Space



Low-dimension space
by using projection



Low-dimension space
by unrolling (2D manifold)

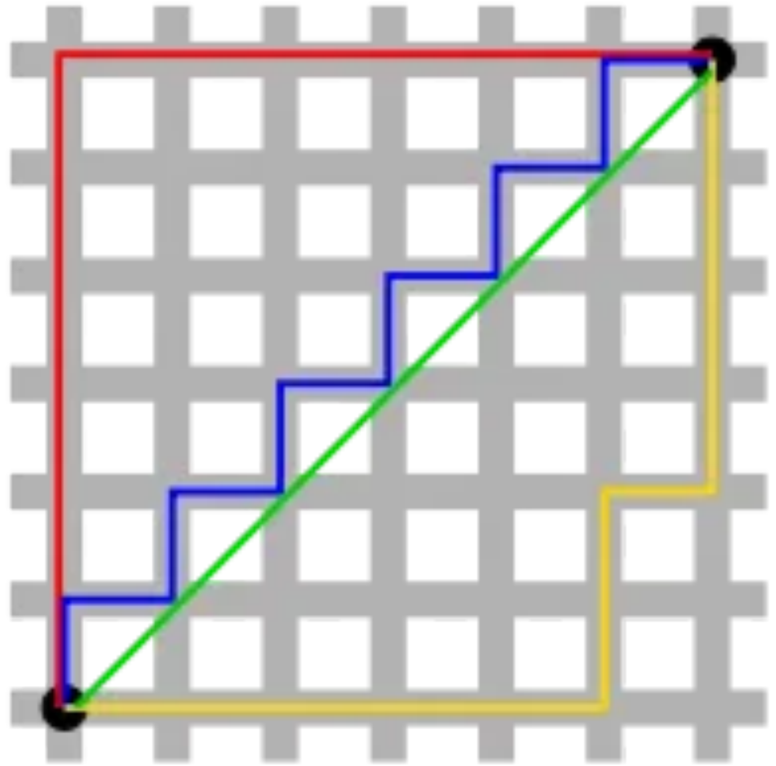
Proximity Matrix

- Only need pair-wise dissimilarity of objects as the input -- Interval scale or Ordinal scale
- Let $\boldsymbol{\Delta}$ be the proximity matrix with entries δ_{ij} representing dissimilarity between two objects i and j.
- Ex: Common dissimilarity metrics:

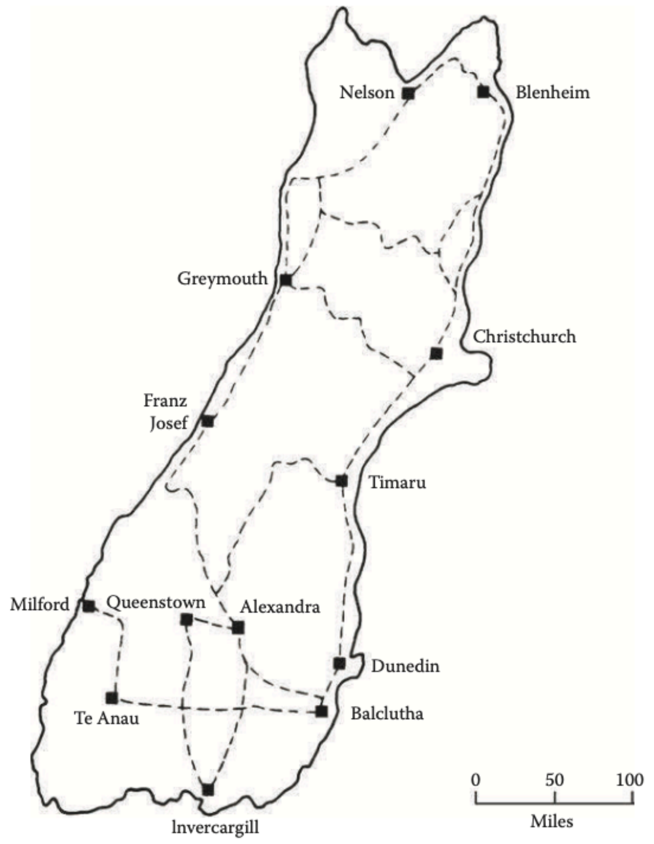
Euclidean Distance: $$\delta_{ij} = \|\boldsymbol{x}_i - \boldsymbol{x}_j\| = \left[\sum_k (x_{ik} - x_{jk})^2\right]^{\frac{1}{2}}$$

Manhattan Distance: $$\delta_{ij} = \sum_k |x_{ik} - x_{jk}|$$

Minowski Distance: $$\delta_{ij} = \left[\sum_k |x_{ik} - x_{jk}|^m\right]^{\frac{1}{m}}$$



- Geographical distance or ranking also possible.



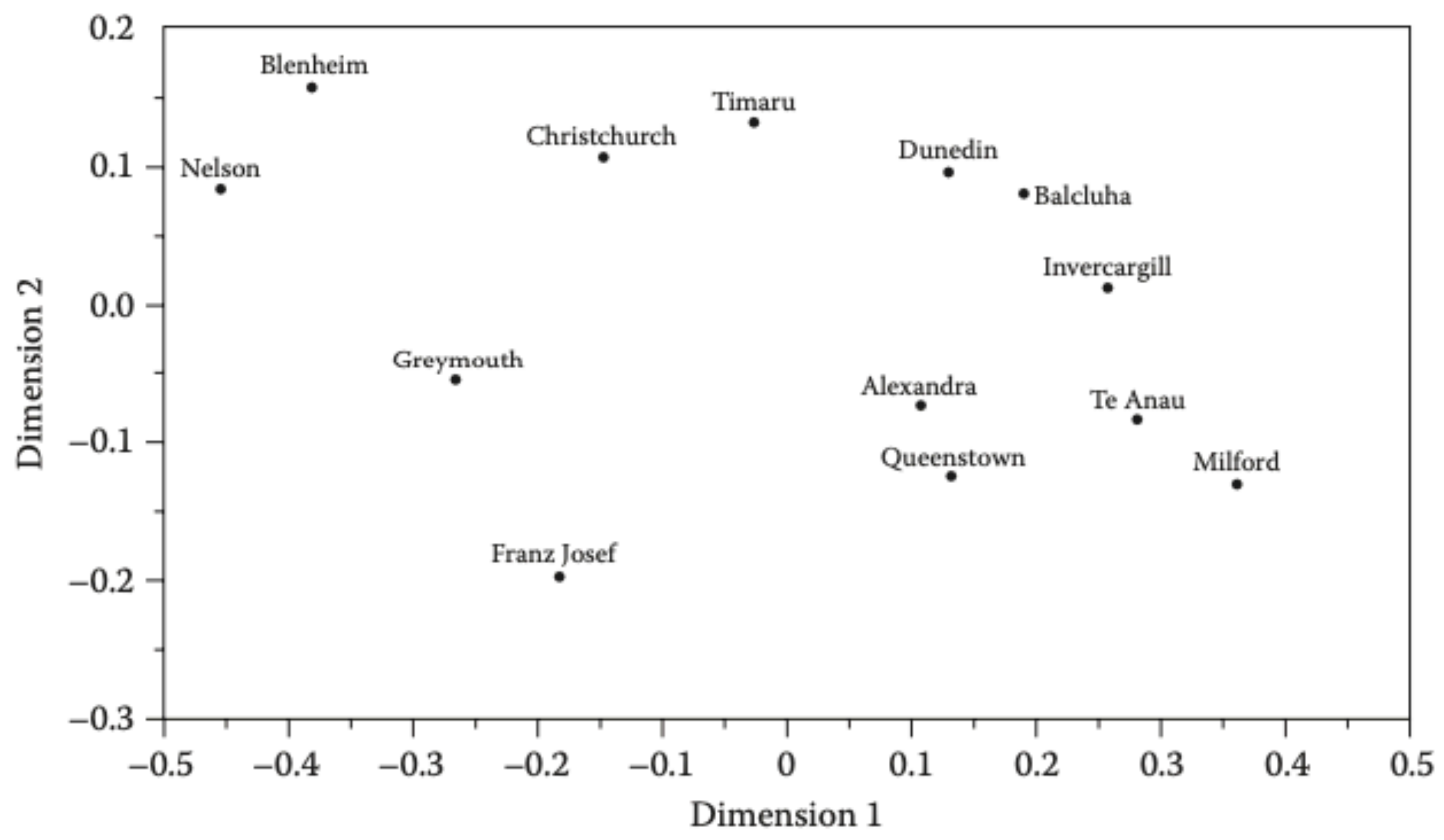
Multi-Dimensional Scaling (MDS)

- Project p-dimensional data to q-dimensional data points $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n \in R^q$ (*principal coordinates* or *configuration*) such that the ***monotonicity*** of the pairwise distances in the original data are preserved in q-space.
- For $M = \dfrac{n(n-1)}{2}$ pair-wise distances descendingly sorted:

$$\delta_{i1,k1} > \delta_{i2,k2} > \cdots > \delta_{in,kn}$$

- Find q-dimensional data points $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n]$, $\mathbf{y}_i \in R^q$, $q \ll p$ such that

$$d_{i1,k1} > d_{i2,k2} > \cdots > d_{in,kn}$$




```
class sklearn.manifold.MDS(n_components=2, *, metric=True, n_init=4,
max_iter=300, verbose=0, eps=0.001, n_jobs=None, random_state=None,
dissimilarity='euclidean', normalized_stress='auto')
```

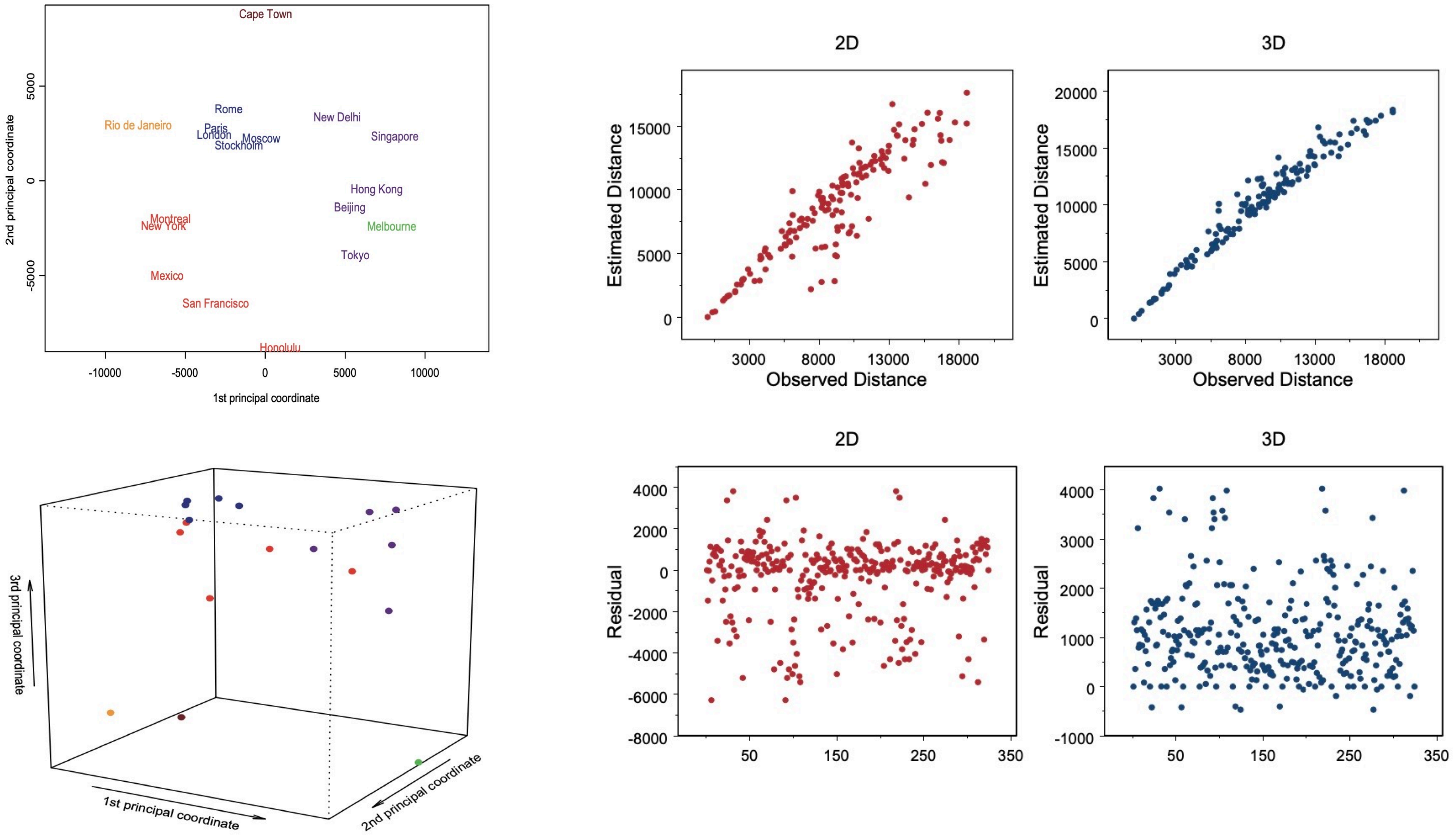
Example: Airline Distance

18 cities

	Beijing	Cape Town	Hong Kong	Honolulu	London	Melbourne
Cape Town	12947					
Hong Kong	1972	11867				
Honolulu	8171	18562	8945			
London	8160	9635	9646	11653		
Melbourne	9093	10338	7392	8862	16902	
Mexico	12478	13703	14155	6098	8947	13557
Montreal	10490	12744	12462	7915	5240	16730
Moscow	5809	10101	7158	11342	2506	14418
New Delhi	3788	9284	3770	11930	6724	10192
New York	11012	12551	12984	7996	5586	16671
Paris	8236	9307	9650	11988	341	16793
Rio de Janeiro	17325	6075	17710	13343	9254	13227
Rome	8144	8417	9300	12936	1434	15987
San Francisco	9524	16487	11121	3857	8640	12644
Singapore	4465	9671	2575	10824	10860	6050
Stockholm	6725	10334	8243	11059	1436	15593
Tokyo	2104	14737	2893	6208	9585	8159

	Mexico	Montreal	Moscow	New Delhi	New York	Paris
Montreal	3728					
Moscow	10740	7077				
New Delhi	14679	11286	4349			
New York	3362	533	7530	11779		
Paris	9213	5522	2492	6601	5851	
Rio	7669	8175	11529	14080	7729	9146
Rome	10260	6601	2378	5929	6907	1108
S.F.	3038	4092	9469	12380	4140	8975
Singapore	16623	14816	8426	4142	15349	10743
Stockholm	9603	5900	1231	5579	6336	1546
Tokyo	11319	10409	7502	5857	10870	9738

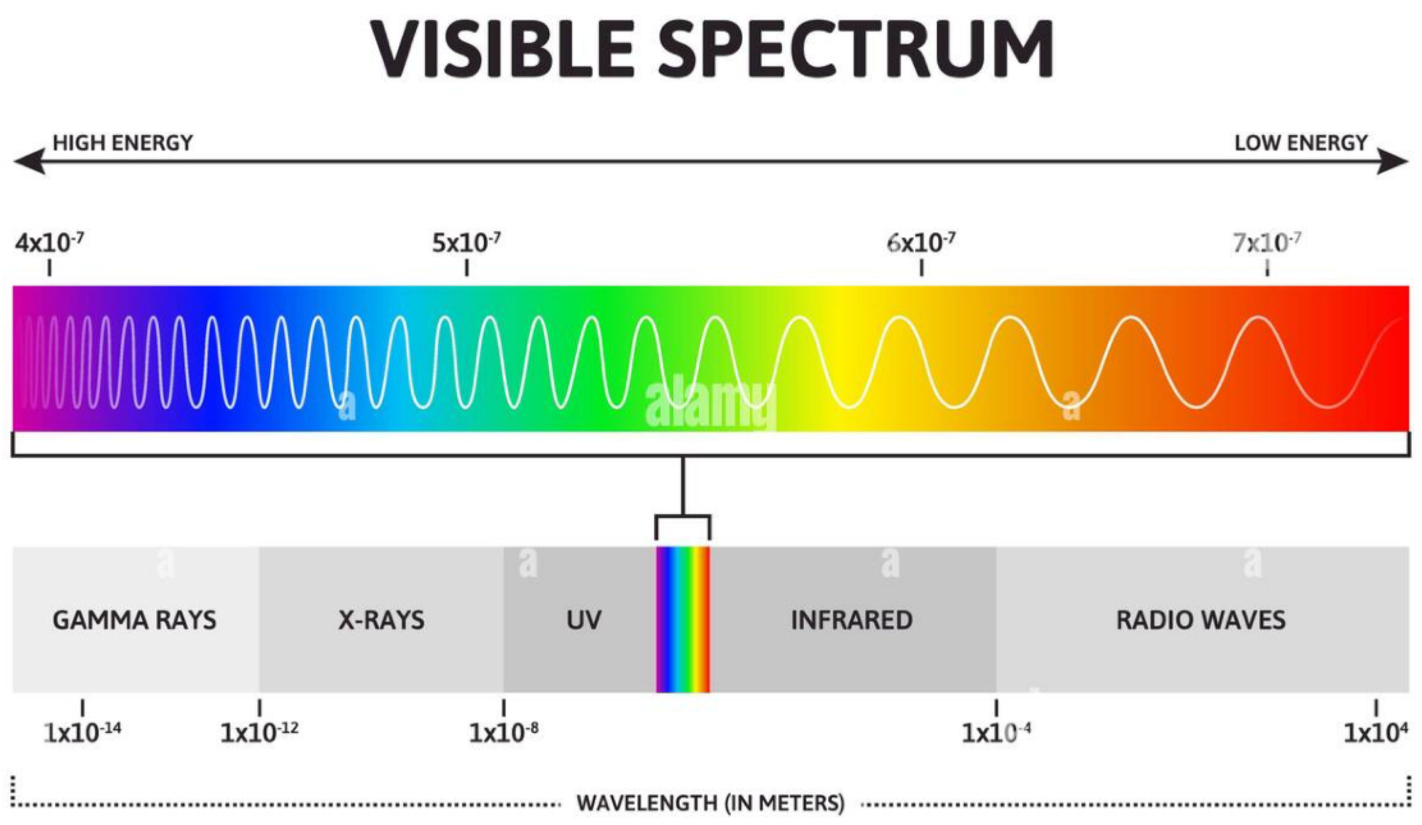
	Rio	Rome	S.F.	Singapore	Stockholm
Rome	9181				
S.F.	10647	10071			
Singapore	15740	10030	13598		
Stockholm	10682	1977	8644	9646	
Tokyo	18557	9881	8284	5317	8193


Cape Town
Rio de Janeiro
Rome
Paris
London
Moscow
Stockholm
New Delhi
Singapore
Hong Kong
Beijing
Melbourne
Montreal
New York
Tokyo
Mexico
San Francisco
Honolulu
2nd principal coordinate
1st principal coordinate
-10000
-5000
0
5000
10000
3rd principal coordinate
1st principal coordinate
2nd principal coordinate
2D
3D
Estimated Distance
Observed Distance
3000
8000
13000
18000
15000
20000
Residual
4000
2000
-2000
-4000
-6000
-8000
3000
1000
-1000
50
150
250
350

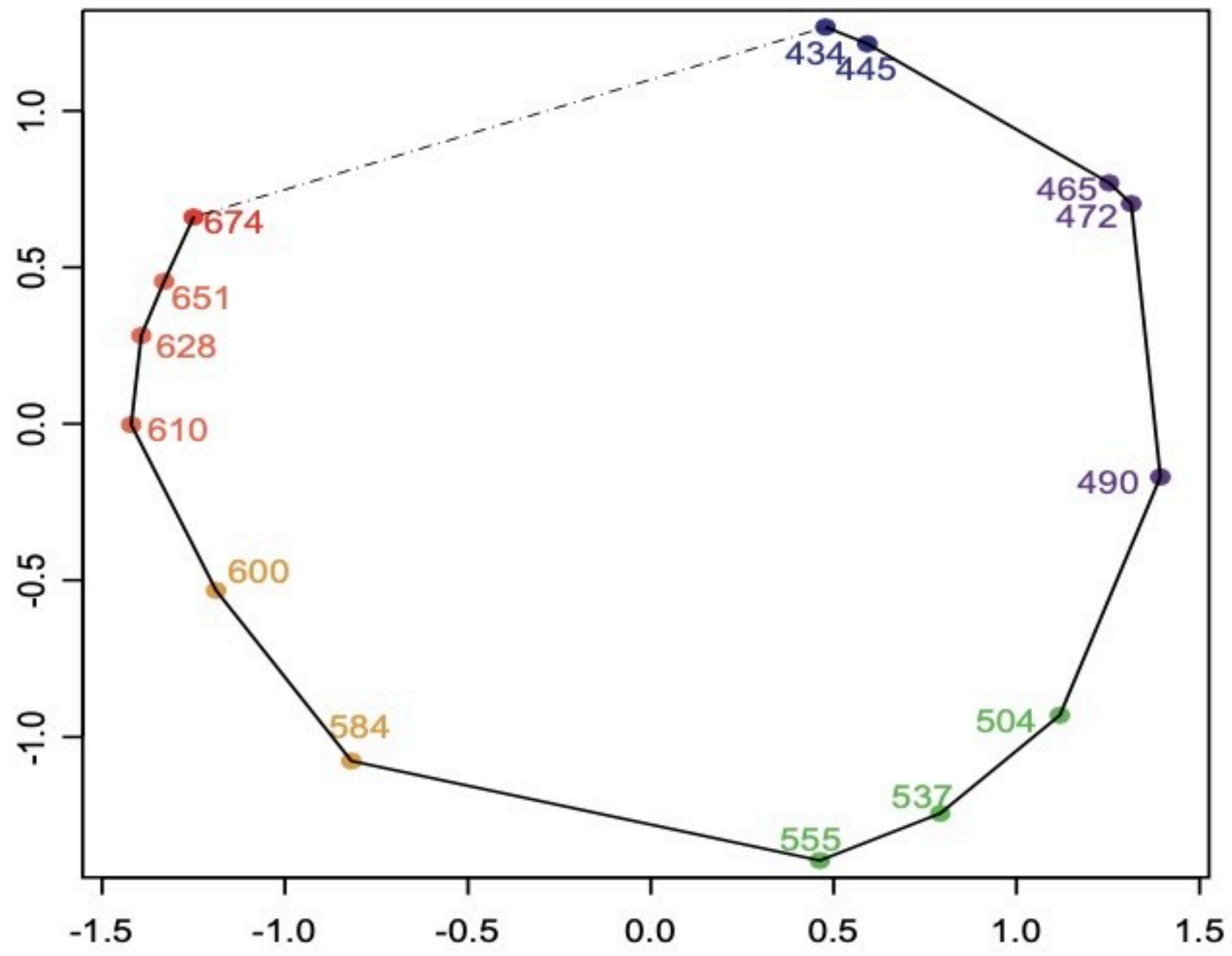
Example: Perceptions of Color Human Vision

	434	445	465	472	490	504	537	555	584	600	610	628	651
445	0.14												
465	0.58	0.50											
472	0.58	0.56	0.19										
490	0.82	0.78	0.53	0.46									
504	0.94	0.91	0.83	0.75	0.39								
537	0.93	0.93	0.90	0.90	0.69	0.38							
555	0.96	0.93	0.92	0.91	0.74	0.55	0.27						
584	0.98	0.98	0.98	0.98	0.93	0.86	0.78	0.67					
600	0.93	0.96	0.99	0.99	0.98	0.92	0.86	0.81	0.42				
610	0.91	0.93	0.98	1.00	0.98	0.98	0.95	0.96	0.63	0.26			
628	0.88	0.89	0.99	0.99	0.99	0.98	0.98	0.97	0.73	0.50	0.24		
651	0.87	0.87	0.95	0.98	0.98	0.98	0.98	0.98	0.80	0.59	0.38	0.15	
674	0.84	0.86	0.97	0.96	1.00	0.99	1.00	0.98	0.77	0.72	0.45	0.32	0.24

Dissimilarity matrix of color wavelengths



Meaningful visual representation of human perception of colors



Color circle

Locally Linear Embedding (LLE)

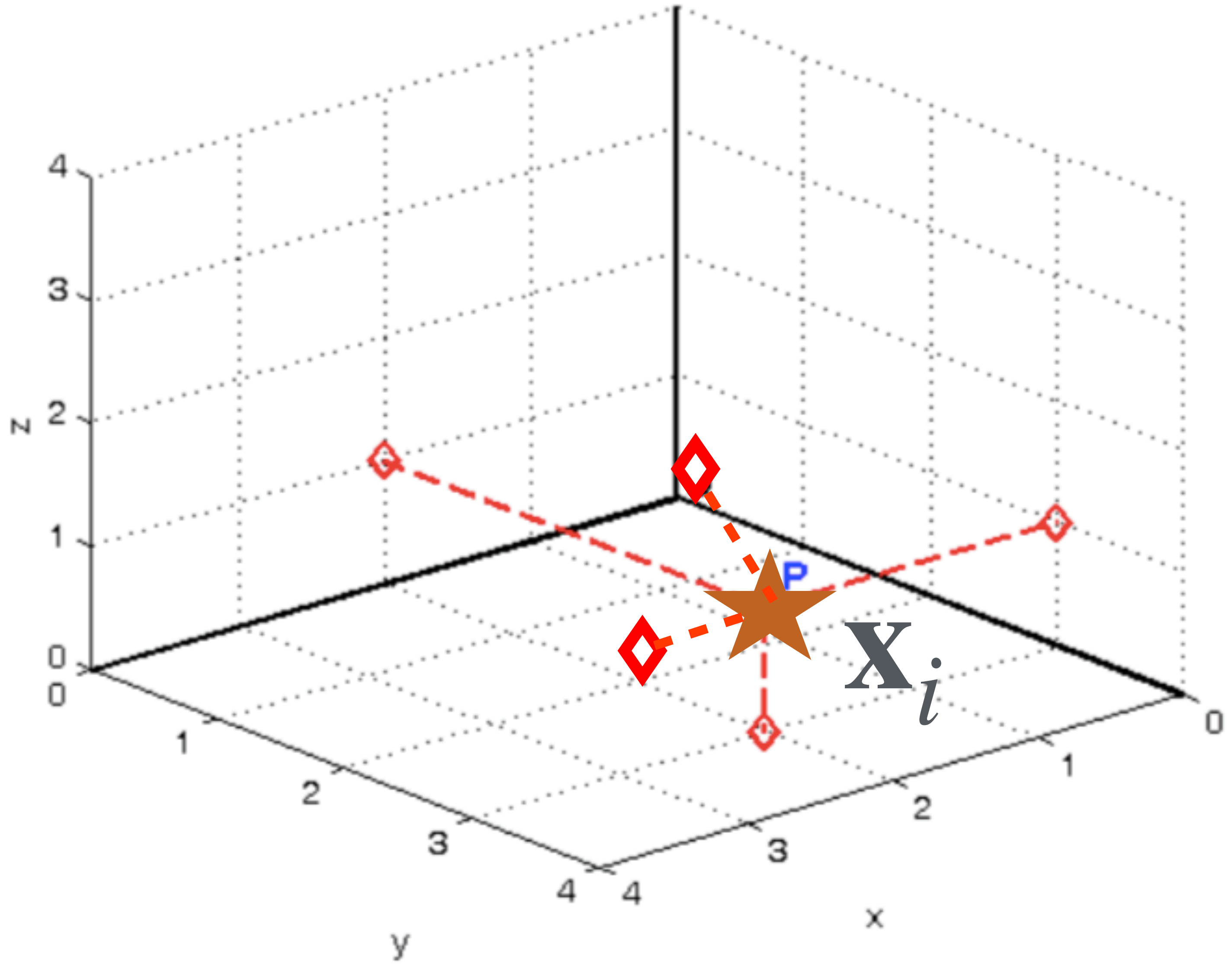
Analyzing overlapping local neighborhoods in order to determine the local structure.

Construct a "neighborhood preserving embedding" of the data into the lower-dimensional space.

1: Find the nearest neighbors of each data point.
2: Express each point $\mathbf{x}_i$ as a linear combination of the other points, i.e., $\mathbf{x}_i = \sum_j w_{ij}\mathbf{x}_j$, where $\sum_j w_{ij} = 1$ and $w_{ij} = 0$ if $\mathbf{x}_j$ is not a near neighbor of $\mathbf{x}_i$.
3: Find the coordinates of each point in lower-dimensional space of specified dimension p by using the weights found in step 2.

Define the weight matrix $\mathbf{W} = [w_{ij}]$, which is obtained by minimizing

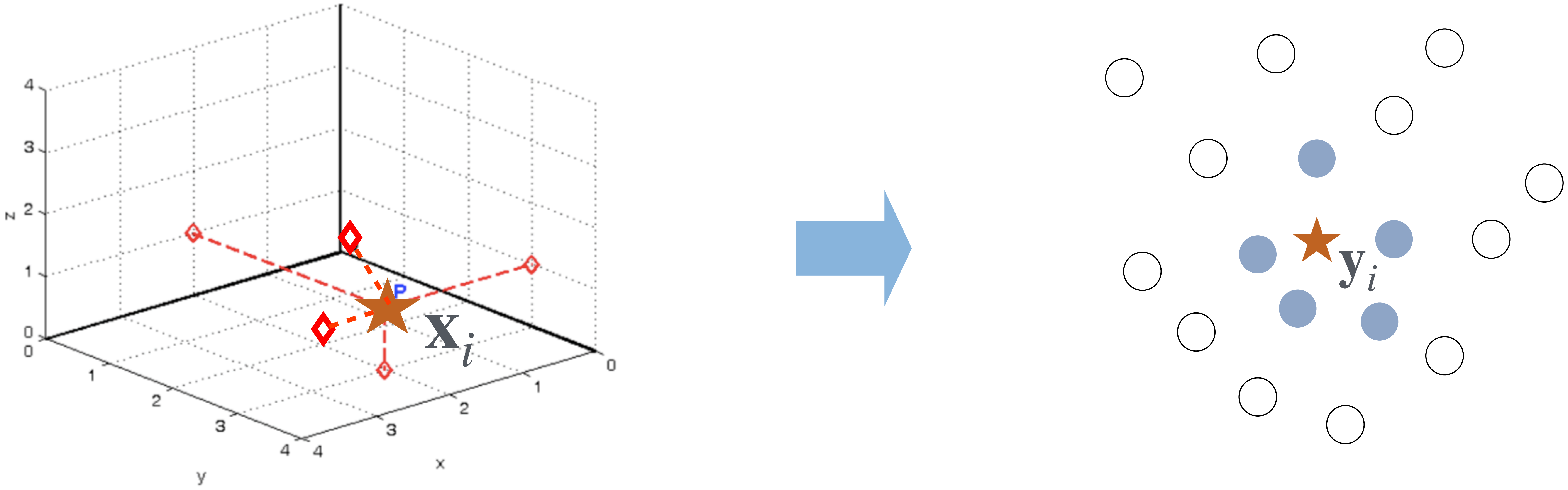
$$\mathrm{Error}(\boldsymbol{W}) = \sum_{i=1}^{n} \left(\boldsymbol{x}_i - \sum_{j \in N(\boldsymbol{x}_i)} w_{ij} \boldsymbol{x}_j \right)^2$$



Let $\mathbf{y}_i$ be a new vector in the reduced dimension r and $\mathbf{Y}$ be the matrix whose i^{th} row is $\mathbf{y}_i$

Given the weight matrix $\mathbf{W}$ and low-dimension r specified by the user, find $\mathbf{Y}$ that minimizes

$$\text{Error}(\boldsymbol{Y}) = \sum_i \left(\boldsymbol{y}_i - \sum_{j \in N(\boldsymbol{x}_i)} w_{ij} \boldsymbol{y}_j \right)^2$$

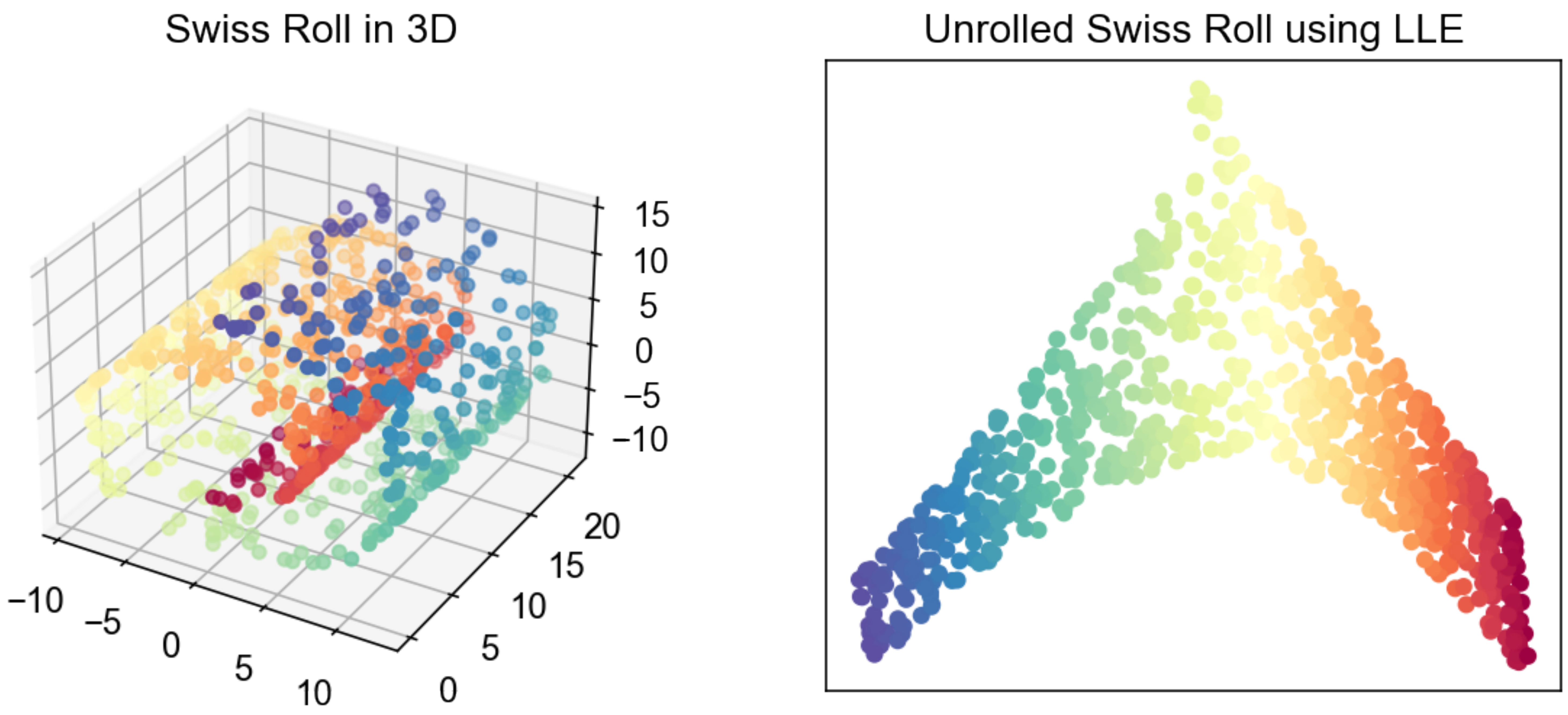


```python
from sklearn.datasets import make_swiss_roll
from sklearn.manifold import LocallyLinearEmbedding

X, color = make_swiss_roll(n_samples=800, random_state=123)
lle = LocallyLinearEmbedding(n_components=2,
                             n_neighbors=10)
X_reduced = lle.fit_transform(X)
```

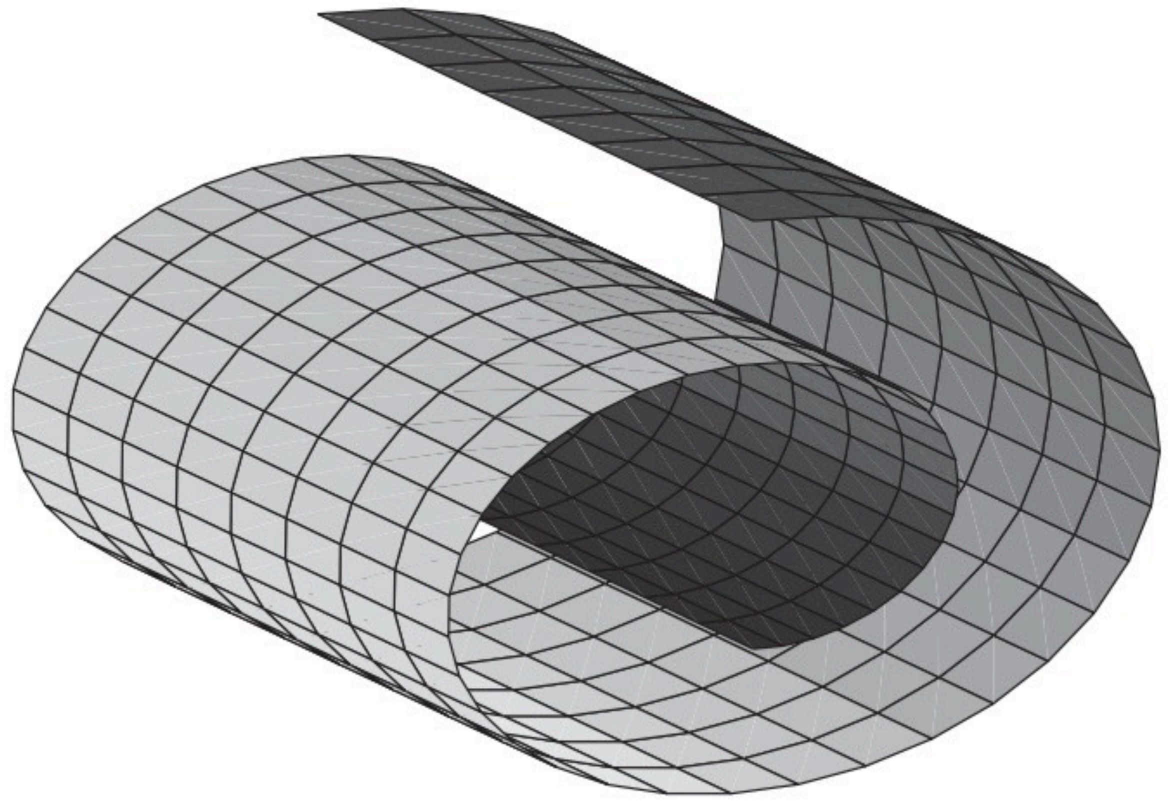
LocallyLinearEmbedding

class sklearn.manifold.**LocallyLinearEmbedding**(*, ***n_neighbors**=5*, ***n_components**=2*, ***reg**=0.001*, ***eigen_solver**='auto'*, ***tol**=1e-06*, ***max_iter**=100*, ***method**='standard'*, ***hessian_tol**=0.0001*, ***modified_tol**=1e-12*, ***neighbors_algorithm**='auto'*, ***random_state**=None*, ***n_jobs**=None*) # [source]



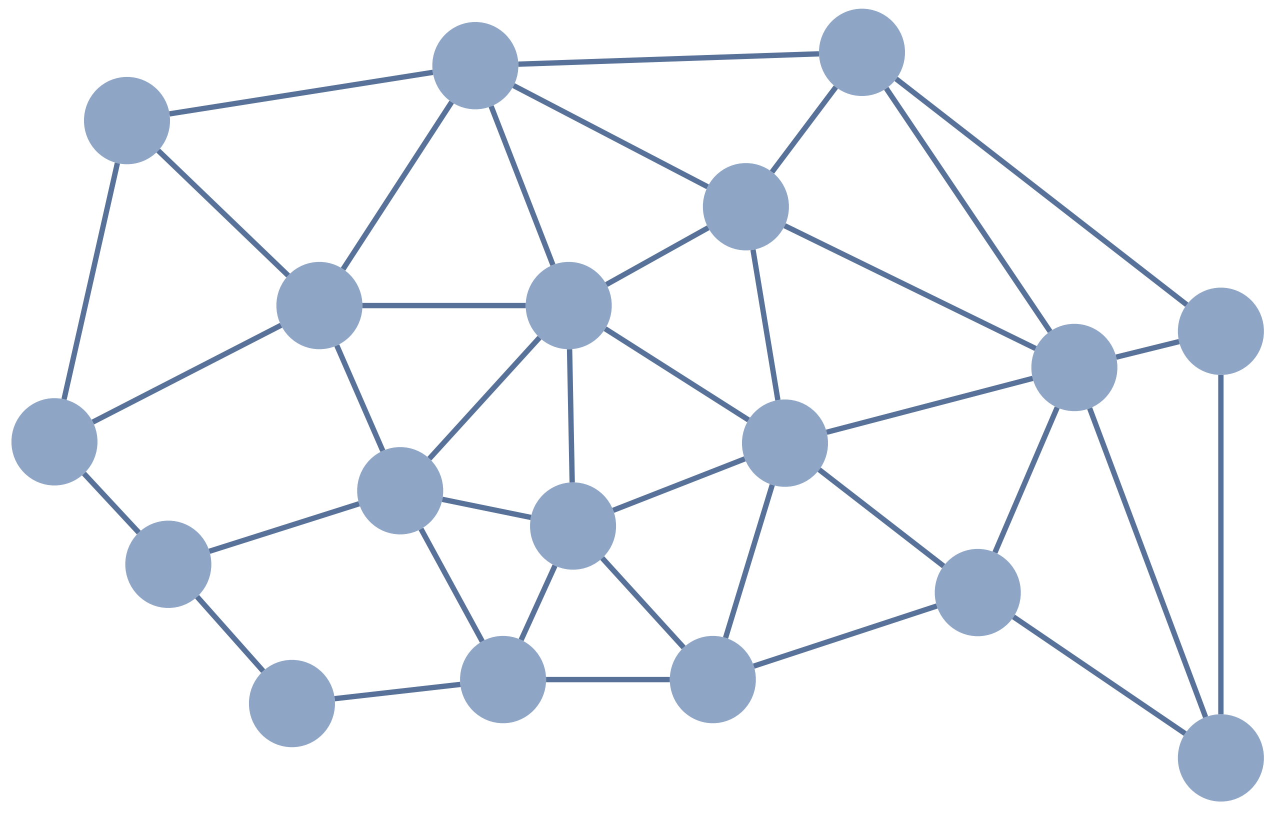
Isometric Feature Mapping (ISOMAP)

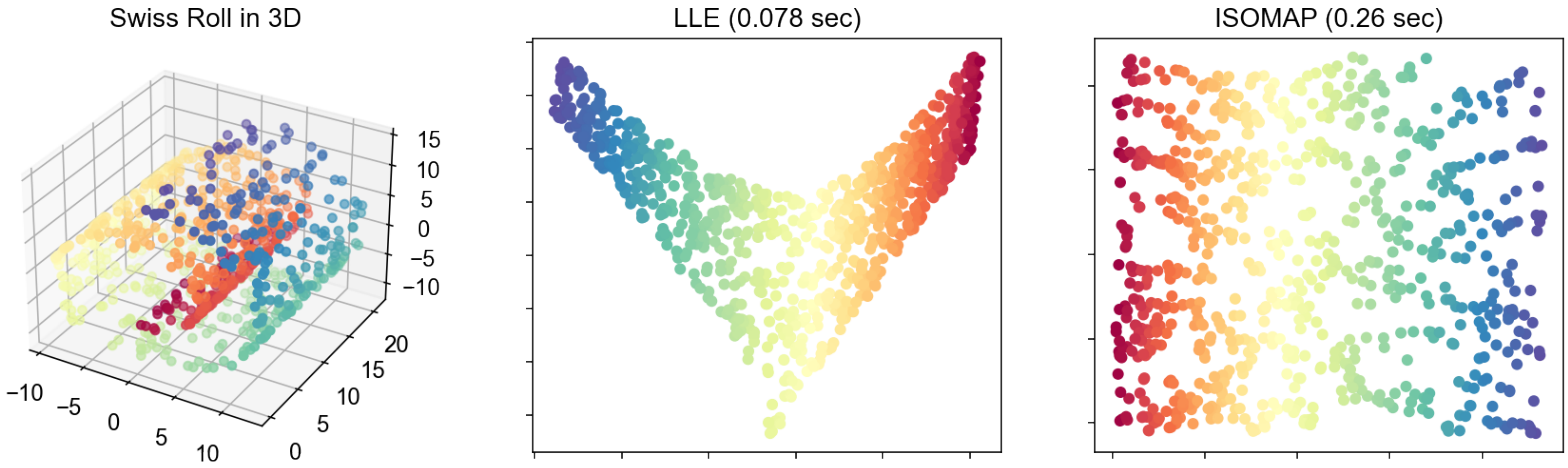
- Extend MDS to handle non-linear relationship in data
- Use ***geodesic distances*** in the original space.



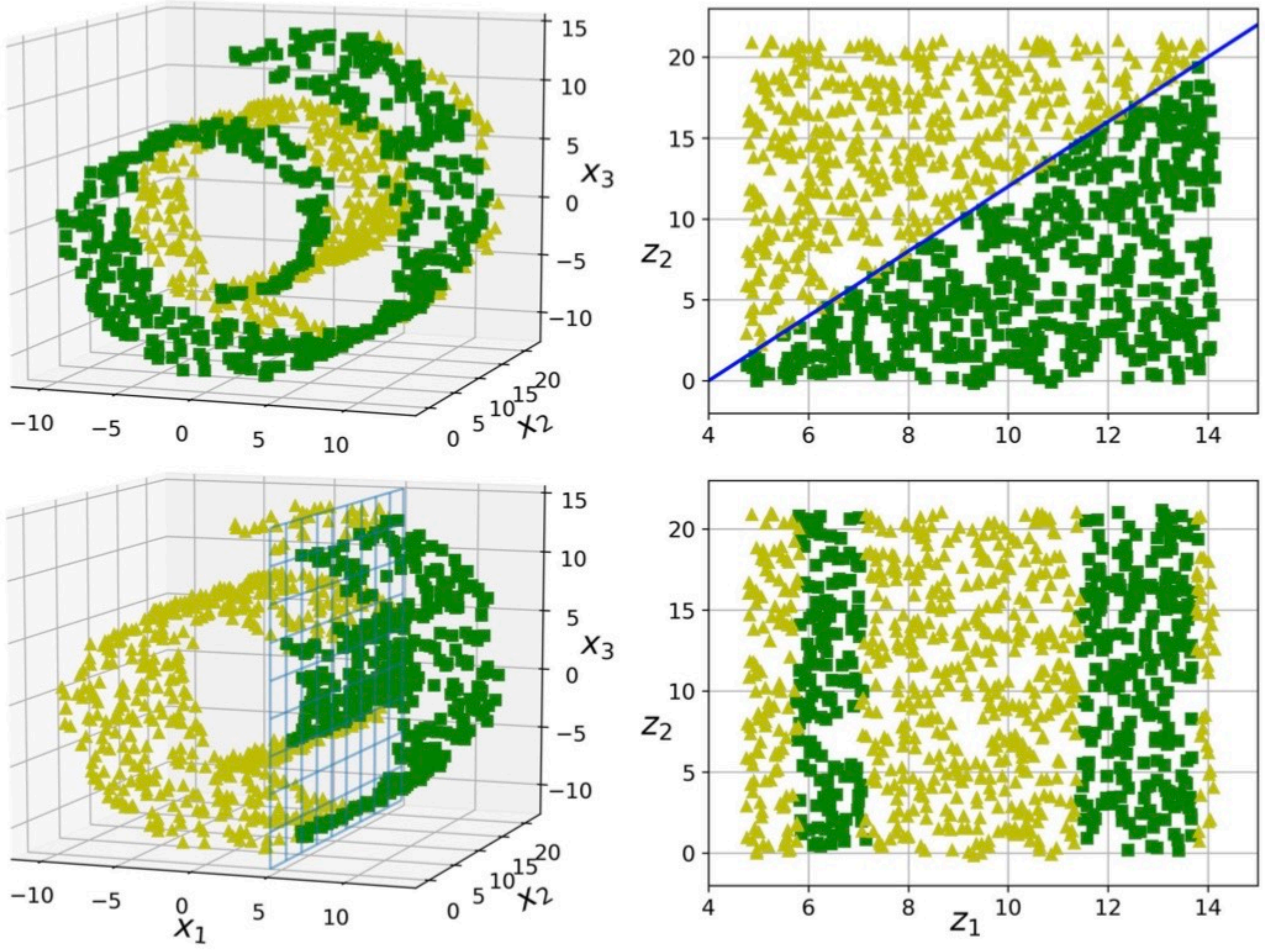
1: Find the nearest neighbors of each data point and create a weighted graph by connecting a point to its nearest neighbors. The nodes are the data points and the weights of the links are the distances between points.
2: Redefine the distances between points to be the length of the shortest path between the two points in the neighborhood graph.
3: Apply classical MDS to the new distance matrix.

Neighborhood graph





Swiss Roll in 3D
LLE (0.078 sec)
ISOMAP (0.26 sec)
−10
−5
0
5
10
0
5
10
15
20
15
10
5
0
−5
−10



Simplified class separation

More complex class separation

Summary

- Dimensionality reduction transforms the dataset into lower dimensions to capture most information of the data.
- PCA projects data into uncorrelated axes with most variances first.
- Manifold learning maps the data into a lower dimension that preserves the distance relationship among points in the original space.
  - Linear manifold learning uses direct pairwise distances in original space.
  - Non-linear manifold learning uses geodesic distance or gives more weights to closer neighbors in original space.

PCA Lab

(Group of 4 - 6)